Classical training of conditional instantaneous quantum polynomial-time Born machines

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Instantaneous quantum polynomial-time (IQP) Born machines support classical training through efficient estimation of the observables required by certain loss functions, such as the maximum mean discrepancy (MMD). However, their current formulation is limited to modeling individual distributions. In this work, we introduce a conditional formulation that encodes the target condition through an input state and jointly trains a shared IQP unitary across a family of distributions. Notably, the encoder need not have an IQP form, provided that the reference state entering the IQP unitary supports efficient classical sampling and amplitude-ratio evaluation. We adapt established amplitude-ratio estimators to obtain unbiased classical estimates of a conditional MMD objective and its gradients. Furthermore, we show how this framework can be used to construct input states that enforce output bit-flip symmetries. Through numerical experiments, we demonstrate that conditional IQP models can learn to control the mean and width of discrete Gaussian distributions, including at conditions not encountered during training. We also show that these models can capture the temperature-dependent transition from unimodal to bimodal magnetization profiles in the Curie–Weiss spin model. Furthermore, we demonstrate the scalability of our approach by training models with up to 2048 qubits. More broadly, our work demonstrates how IQP Born machines can be generalized to conditional generation tasks, expanding the range of problems that can be addressed using classically trainable IQP circuits.

I. INTRODUCTION

Conditional generative models approximate a family of target distributions $p(\cdot \mid c)$ using a single model. They capture how a distribution depends on one or more conditioning variables, such as temperature in a physical system or a class label in image generation. Learning across conditions aims to capture structures common to the family, which in turn supports generation at conditions not encountered during training. Classical examples include conditional variational autoencoders [1–3], generative adversarial networks [4–6], and diffusion models [7, 8]. In the quantum setting, one of the most common architectures is the quantum circuit Born machine (QCBM), which represents probability distributions through its measurement statistics [9]. QCBMs have been applied to conditional generation tasks through various mechanisms, including input-state preparation, conditional registers, and postselection [10–12].

Instantaneous quantum polynomial-time (IQP) circuits offer a particularly attractive design for QCBMs, as certain families are conjectured to be classically hard to sample from under standard complexity-theoretic assumptions [13, 14]. Nevertheless, classical algorithms exist that can efficiently estimate their Pauli-Z expectation values to additive accuracy [15]. This distinction enables the classical training of IQP Born machines using objective functions such as the maximum mean discrepancy (MMD), which can be expressed in terms of these expectation values [16, 17]. Consequently, we can classically estimate the objective function and its gradients, while quantum hardware is required for sampling [18].

Recent efforts in the classical training of IQP circuits have explored how to introduce generative control by adapting components of the generator. For example, ConQuER fits an additional controller to data filtered by a target condition while keeping a pretrained generator fixed [19]. Similarly, Liu *et al.* retain a shared circuit, fit a small parameter block separately for selected conditions, and interpolate the fitted parameters to generate samples at unseen conditions [20]. Their subsequent work uses physics-informed initialization to improve condition-specific latent adaptation [21]. In contrast, we introduce a classically trainable conditional IQP formulation in which an encoder prepares $|\phi(c)\rangle$ and a shared parameterized unitary defines $q_{\boldsymbol{\theta}}(x \mid c) = |\langle x|U_{\boldsymbol{\theta}}|\phi(c)\rangle|^2$. Notably, the encoder need not have a strict IQP form, provided that its reference state $|\psi_c\rangle = H^{\otimes n}|\phi(c)\rangle$ supports efficient classical sampling and amplitude-ratio evaluation. These operations allow us to apply established sparse-observable estimation methods to obtain unbiased estimates of conditional Pauli-Z expectation values and their phase derivatives [15, 22]. Combining these estimators with the MMD yields unbiased estimates of the conditional training objective and its gradients [17, 18]. Figure 1 summarizes the framework.

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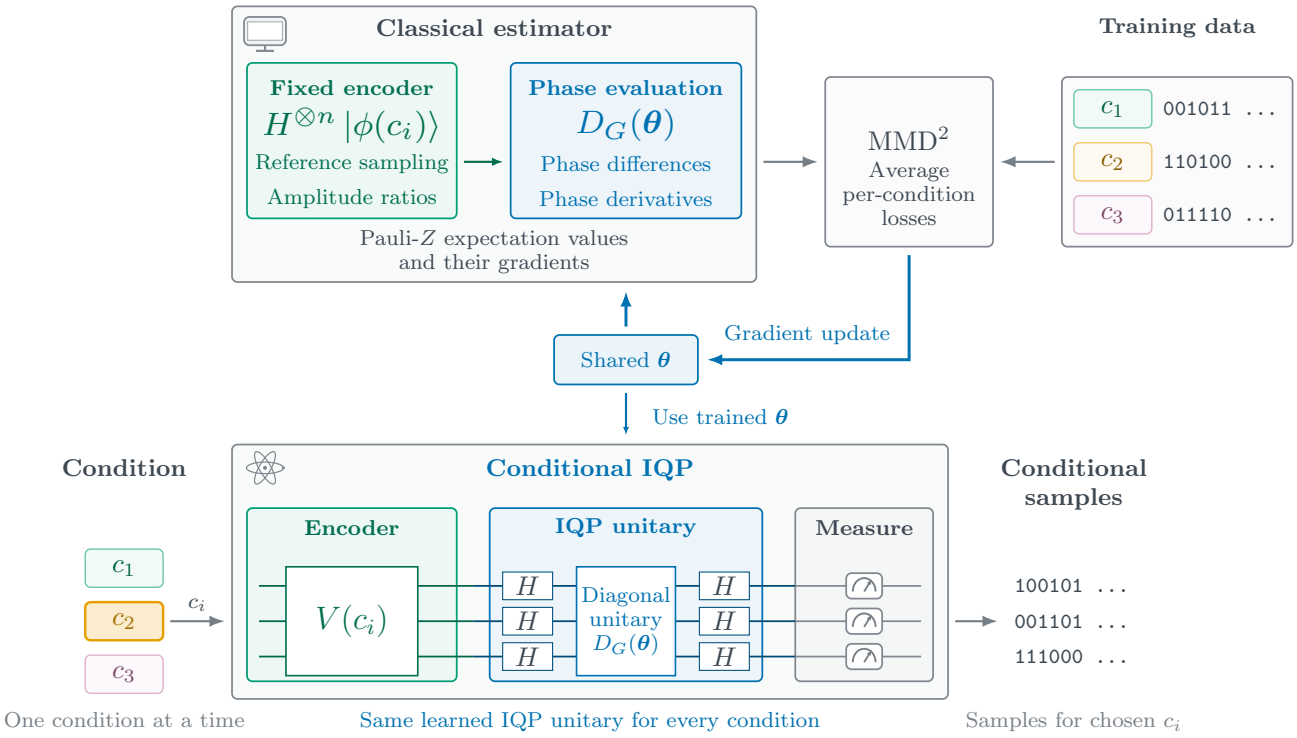


FIG. 1. Proposed framework for classical training of conditional instantaneous quantum polynomial-time (IQP) Born machines. An encoder maps each condition c_i to an input state $|\phi(c_i)\rangle$. During classical training, samples and amplitude ratios from the reference state $H^{\otimes n}|\phi(c_i)\rangle$, together with phase differences induced by the IQP layer, yield estimates of conditional Pauli-Z moments and their gradients. We use these estimates to optimize a joint IQP unitary for all input states with a squared maximum mean discrepancy (MMD) objective averaged across training conditions. During inference, a chosen condition c_i determines the input state. Applying the trained IQP unitary $U_{\boldsymbol{\theta}}$ to the input state $|\phi(c_i)\rangle$ and measuring in the computational basis produces samples from the learned conditional distribution $q_{\boldsymbol{\theta}}(\cdot \mid c_i)$.

We validate the framework through sixteen-qubit numerical experiments. First, we show that conditional IQP models can control the mean of binomial Hamming-weight distributions and extend this task to a two-coordinate family with variable mean and width. Next, we model a Curie–Weiss distribution and demonstrate that conditional IQP models can capture the temperature-dependent transition from unimodal to bimodal magnetization profiles, including at temperatures not encountered during training. We also present a graph-based encoder construction that enforces multiple independent output bit-flip symmetries as an example of utility of this framework beyond conditional generation. Finally, we demonstrate the computational feasibility of our approach through classical training runs with up to 2048 qubits.

The rest of the paper is organized as follows: Sections II and III describe the model and classical training. Section IV reports the experiments, and Section V discusses input-induced symmetries. Sections VI and VII discuss the findings and conclude.

II. CONDITIONAL IQP MODEL

In this section, we present the conditional IQP model formulation, separating condition encoding from the learned unitary transformation. An encoder maps each condition $c \in \mathcal{C}$ to an input state $|\phi(c)\rangle$, and the same parameterized IQP unitary $U_{\boldsymbol{\theta}}$ is applied to every encoded state. For a target family of distributions $\{p(\cdot \mid c) : c \in \mathcal{C}\}$ over n-bit strings, we define the model distribution as

$$q_{\boldsymbol{\theta}}(x \mid c) = \left|\langle x| U_{\boldsymbol{\theta}} |\phi(c)\rangle\right|^2, \qquad x \in \{0,1\}^n. \tag{1}$$

Throughout this work, we use "IQP unitary" to refer exclusively to the learned block $U_{\boldsymbol{\theta}}$. The encoder isn't restricted to an IQP form and may include non-diagonal operations. Consequently, the complete conditional circuit may not belong to the strict IQP circuit class.

We parameterize the shared IQP unitary using an undirected graph $G = (V, E)$ with $n = |V|$ vertices. Each vertex $i \in V$ contributes a one-body generator Z_i, while each edge $\{i, j\} \in E$ contributes a two-body generator $Z_i Z_j$. The unitary is then expressed as

$$U_{\boldsymbol{\theta}} = H^{\otimes n} D_G(\boldsymbol{\theta}) H^{\otimes n}, \tag{2}$$

where

$$D_G(\boldsymbol{\theta}) = \exp\left(i \sum_{i \in V} \theta_i Z_i + i \sum_{\{i,j\} \in E} \theta_{ij} Z_i Z_j \right). \quad (3)$$

Although we restrict the ansatz in our numerical experiments to one- and two-body generators, the formulation naturally extends to higher-order interactions by including additional k-body Pauli-Z generators.

The initial Hadamard layer maps the encoded input to the reference state $|\psi_c\rangle = H^{\otimes n} |\phi(c)\rangle$, immediately before the diagonal IQP layer. We assume that each encoded input is normalized and independent of the learned parameters $\boldsymbol{\theta}$, so the reference state inherits the same properties. We train the model using samples and amplitude ratios from this reference state. Its computational-basis amplitudes are

$$\psi_c(z) = \langle z | \psi_c \rangle = \langle z | H^{\otimes n} |\phi(c)\rangle , \quad (4)$$

The squared amplitudes $|\psi_c(z)|^2$ give the reference sampling distribution for condition c. Our requirements concern classical access to this state rather than the encoder's gate structure. For the relevant conditions c and bit masks a, we require two efficient classical operations:

1. Sample $z \sim |\psi_c(z)|^2$.
2. Evaluate the amplitude ratio

$$R_c(z, a) = \frac{\psi_c(z \oplus a)}{\psi_c(z)} \quad (5)$$

for sampled z, where $\oplus$ denotes bitwise addition modulo two.

We call these two operations the sampling-and-ratio interface.

We estimate Pauli-Z moments by combining reference amplitude ratios with phase factors from the IQP layer. For each sampled string z, the ratio $R_c(z, a)$ compares the reference amplitudes of z and the paired string $z \oplus a$. The IQP layer contributes a complex phase factor determined by the phase difference between these strings. Multiplying the amplitude ratio by this phase factor and averaging the real part over reference samples yields an estimate of the corresponding Pauli-Z moment [15]. Note that, we evaluate amplitude ratios only on the support of $|\psi_c|^2$, since the sampling never selects strings for which $\psi_c(z) = 0$.

Computationally tractable reference states, in the sense of Van den Nest, admit efficient computational-basis sampling and amplitude evaluation [15]. These states provide a sufficient realization of the sampling-and-ratio interface, provided that their amplitudes can be evaluated with sufficient precision to compute the required ratios efficiently. The simplest example is a product-state encoder $|\phi(c)\rangle = \bigotimes_{i=1}^{n} |\phi_i(c)\rangle$. The corresponding reference amplitudes factorize as $\psi_c(z) = \prod_i \psi_{c,i}(z_i)$, where $\psi_{c,i}(b) = \langle b | H |\phi_i(c)\rangle$. Reference samples can therefore be generated by drawing each bit independently, while the amplitude ratio takes the form

$$R_c(z, a) = \prod_{i : a_i = 1} \frac{\psi_{c,i}(1 - z_i)}{\psi_{c,i}(z_i)}. \quad (6)$$

When the local amplitudes can be evaluated at constant arithmetic cost, both operations require at most $O(n)$ local operations. Beyond this product-state example, Section V introduces correlated reference states that satisfy the same access requirements while enforcing exact output symmetries.

However, efficient access to the sampling-and-ratio interface alone does not guarantee that the model can solve a given task, as the encoder also constrains how the output probabilities depend on the condition c. For rotation encodings with angles linear in c, the encoding generators determine the accessible Fourier frequencies, while the learned parameters control their coefficients within the constraints imposed by the circuit architecture [23]. Multidimensional Fourier analyses further characterize how the encoding and circuit architecture constrain the model's expressive capacity [24]. Moreover, Caro et al. derived encoding-dependent generalization bounds for parameterized quantum circuits [25]. Overall, these results highlight that encoder design plays a substantial role and can be as important as the choice of circuit architecture.

Lastly, existing IQP sampling-hardness results apply to specific circuit families and do not establish hardness for arbitrary condition-dependent input states or specific parameter values [13, 26]. Furthermore, to deploy conditional IQP circuits on a quantum computer we also need an efficient implementation for preparing $|\phi(c)\rangle$, which is not guaranteed by the classical sampling-and-ratio interface alone.

A. Conditional Pauli estimator and gradients

In Section III, we express the conditional MMD objective in terms of Pauli-Z moments. For a bit mask $a \in \{0,1\}^n$, we define $Z_a = \bigotimes_{i=1}^{n} Z_i^{a_i}$ and denote its conditional model expectation by $\mu_q(a \mid c)$. We use the sampling-and-ratio interface to derive estimators for these moments and their derivatives with respect to the parameters of the IQP unitary.

Van den Nest established efficient sparse-observable estimation for computationally tractable states [15], while Havlíček developed estimation methods based on sampling and amplitude-ratio access [22]. Recio-Armengol *et al.* applied classical Pauli-moment estimation to MMD training of IQP Born machines [18]. Here, we extend these methods to prescribed condition-dependent reference states, enabling joint training of a single shared IQP unitary across conditions.

We denote the diagonal IQP layer by $D_{\boldsymbol{\theta}}$ and express its action as

$$D_{\boldsymbol{\theta}} |z\rangle = e^{i f_{\boldsymbol{\theta}}(z)} |z\rangle , \tag{7}$$

where

$$f_{\boldsymbol{\theta}}(z) = \sum_{j=1}^{P} \theta_j \chi_{g_j}(z), \qquad \chi_{g_j}(z) = (-1)^{g_j \cdot z}, \tag{8}$$

P denotes the number of generators, and $g_j \in \{0,1\}^n$ specifies the support of the jth Pauli-Z generator. The dot product $g_j \cdot z$ is evaluated modulo two. We define the phase difference associated with a bit mask a as

$$\Delta_a f_{\boldsymbol{\theta}}(z) = f_{\boldsymbol{\theta}}(z \oplus a) - f_{\boldsymbol{\theta}}(z). \tag{9}$$

The final Hadamard layer transforms Z_a into $X_a = \bigotimes_i X_i^{a_i}$. Consequently, the conditional Pauli moment can be expressed as

$$\begin{aligned} \mu_q(a \mid c) &= \langle \psi_c | D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}} |\psi_c\rangle \\ &= \sum_z \psi_c(z)^* \psi_c(z \oplus a) \\ &\quad \times e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}. \end{aligned} \tag{10}$$

Terms with $\psi_c(z) = 0$ vanish. On the support of the reference sampling distribution, we can write

$$\psi_c(z)^* \psi_c(z \oplus a) = |\psi_c(z)|^2 R_c(z,a). \tag{11}$$

The sum therefore becomes an expectation over reference samples, with each amplitude ratio multiplied by the corresponding IQP phase factor. Since the Pauli moment is real, we take the real part of this product.

For a reference sample $z \sim |\psi_c|^2$, we define

$$\begin{aligned} W_{c,a}(z;\boldsymbol{\theta}) &= R_c(z,a) e^{i \Delta_a f_{\boldsymbol{\theta}}(z)}, \\ Y_{c,a}(z;\boldsymbol{\theta}) &= \mathrm{Re}\, W_{c,a}(z;\boldsymbol{\theta}). \end{aligned} \tag{12}$$

The derivative of the phase difference with respect to θ_j is

$$\begin{aligned} d_{j,a}(z) &= \partial_{\theta_j} \Delta_a f_{\boldsymbol{\theta}}(z) \\ &= \chi_{g_j}(z \oplus a) - \chi_{g_j}(z) \in \{-2, 0, 2\}. \end{aligned} \tag{13}$$

For a fixed condition c and bit mask a, we assume that the reference state is normalized, independent of $\boldsymbol{\theta}$, and supports the sampling-and-ratio interface. For each reference sample, we define the phase-gradient estimator as

$$Y'_{j,c,a}(z;\boldsymbol{\theta}) = \mathrm{Re}[i\, d_{j,a}(z) W_{c,a}(z;\boldsymbol{\theta})]\,. \tag{14}$$

We also define the operators

$$\begin{aligned} A_a &= D_{\boldsymbol{\theta}}^\dagger X_a D_{\boldsymbol{\theta}}, \\ B_{ja} &= \partial_{\theta_j} A_a = i[A_a, Z_{g_j}]. \end{aligned} \tag{15}$$

Both operators are Hermitian and have at most one nonzero entry per row. The real parts of their local sparse-observable estimators are $Y_{c,a}$ and $Y'_{j,c,a}$, respectively. Since the reference state is independent of $\boldsymbol{\theta}$, these estimators satisfy, under exact sampling and arithmetic,

$$\begin{aligned} \mathbb{E}[Y_{c,a}] &= \mu_q(a \mid c), \\ \mathbb{E}[Y'_{j,c,a}] &= \partial_{\theta_j} \mu_q(a \mid c). \end{aligned} \tag{16}$$

The operator norm bounds $\|A_a\| = 1$ and $\|B_{ja}\| \le 2$ imply second moments bounded by one and four, respectively [22, Sec. 3.4]. We evaluate amplitude ratios only at sampled points with nonzero reference amplitudes. In particular,

$$\mathbb{E}[|W_{c,a}|^2] = \sum_{z:\psi_c(z)\neq 0} |\psi_c(z \oplus a)|^2 \le 1. \tag{17}$$

Averaging S independent samples therefore yields mean-square errors of at most $1/S$ and $4/S$ for a given moment and phase derivative, respectively. By Chebyshev's inequality, $O(\epsilon^{-2})$ samples suffice to achieve additive error ϵ with fixed confidence.

In practice, we draw S independent reference samples $z_s \sim |\psi_c|^2$ and estimate the moment as

$$\widehat{\mu}_q(a \mid c) = \frac{1}{S} \sum_{s=1}^{S} Y_{c,a}(z_s;\boldsymbol{\theta}). \tag{18}$$

Similarly, we estimate the phase derivative by averaging the corresponding $Y'_{j,c,a}$ values over the same samples. For the standard IQP input $|0\rangle^{\otimes n}$, all reference amplitudes are equal, giving $R_c(z,a) = 1$. The moment estimator then reduces to $Y = \cos[\Delta_a f_{\boldsymbol{\theta}}(z)]$, recovering the usual IQP cosine estimator [15, 27].

III. CONDITIONAL IQP TRAINING

Section II provides classical estimators for conditional Pauli-Z moments and their gradients. We now combine these estimators with the moment-based formulation of the Gaussian-kernel MMD to train conditional IQP circuits. Throughout this work, we use *classically trainable* to mean that the training objective and its gradients can be efficiently estimated to a prescribed additive accuracy on a classical computer.

A. Condition-balanced MMD objective

We optimize the shared parameter vector $\boldsymbol{\theta}$ by minimizing the squared maximum mean discrepancy (MMD) across the training conditions. For distributions p and q on $\{0,1\}^n$ and a positive-definite kernel k, the population squared MMD is [16]

$$\begin{aligned} \mathrm{MMD}^2(p,q) = \mathbb{E}_{x,x'\sim p}[k(x,x')] - 2\mathbb{E}_{x\sim p,\, y\sim q}[k(x,y)] \\ + \mathbb{E}_{y,y'\sim q}[k(y,y')], \end{aligned} \tag{19}$$

where we draw the two samples in each expectation independently.

Throughout this work, we use the Gaussian kernel,

$$k_\sigma(x,y) = \exp\left[-\frac{d_H(x,y)}{2\sigma^2}\right], \tag{20}$$

Here, $d_H(x,y)$ denotes the Hamming distance, which counts the positions at which x and y differ, and $\sigma > 0$ is the kernel bandwidth. For binary strings, $d_H(x,y) = \|x-y\|_2^2$, so the kernel corresponds to the standard Gaussian kernel restricted to the binary domain. For any finite $\sigma > 0$, the MMD vanishes if and only if $p = q$ [16].

Let $\mathcal{C}_{\text{train}}$ denote the set of training conditions and τ the set of kernel bandwidths. We train the shared model by minimizing the condition-balanced objective

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{C}_{\text{train}}|}\sum_{c\in\mathcal{C}_{\text{train}}}\frac{1}{|\tau|}\sum_{\sigma\in\tau}\text{MMD}_\sigma^2(p(\cdot\mid c), q_{\boldsymbol{\theta}}(\cdot\mid c))\,. \tag{21}$$

Here, MMD_σ^2 denotes the MMD computed with the kernel k_σ. We weight all training conditions and bandwidths equally. At each optimizer update, we independently draw one condition uniformly from $\mathcal{C}_{\text{train}}$ and one bandwidth uniformly from τ, giving a stochastic estimate of this average.

B. Character expansion

We use the Gaussian-kernel expansion into binary characters employed in earlier QCBM and classical IQP training work [17, 18]. We apply this expansion separately at each condition to express the shared training objective through conditional Pauli-Z moments. We sample a bit mask $a \in \{0,1\}^n$ by drawing its coordinates independently according to

$$a_i \sim \text{Bernoulli}(p_\sigma), \qquad p_\sigma = \frac{1-e^{-1/(2\sigma^2)}}{2}, \tag{22}$$

The corresponding binary character is

$$\chi_a(x) = (-1)^{a\cdot x}, \tag{23}$$

which records the parity of the bits selected by a. For each bit where x and y agree, the expected character-product factor is 1. Where they differ, it is $1-2p_\sigma = e^{-1/(2\sigma^2)}$. Since the mask coordinates are independent, multiplying these factors recovers the Gaussian kernel:

$$k_\sigma(x,y) = \mathbb{E}_a[\chi_a(x)\chi_a(y)]\,. \tag{24}$$

For a condition c, define the target and model character moments

$$\begin{aligned}\mu_p(a\mid c) &= \mathbb{E}_{x\sim p(\cdot|c)}[\chi_a(x)],\\ \mu_q(a\mid c) &= \mathbb{E}_{x\sim q_{\boldsymbol{\theta}}(\cdot|c)}[\chi_a(x)].\end{aligned} \tag{25}$$

Substituting Eq. (24) into Eq. (19) gives

$$\text{MMD}_\sigma^2(p(\cdot\mid c), q_{\boldsymbol{\theta}}(\cdot\mid c)) = \mathbb{E}_a\Big[(\mu_p(a\mid c)-\mu_q(a\mid c))^2\Big]\,. \tag{26}$$

Thus, the squared MMD averages the squared differences between target and model character moments over the mask distribution.

For a Born machine, the Pauli operator

$$Z_a = \bigotimes_{i=1}^n Z_i^{a_i} \tag{27}$$

has eigenvalue $\chi_a(x)$ on the computational-basis state $|x\rangle$. Its model expectation value is therefore

$$\mu_q(a\mid c) = \langle Z_a\rangle_{q_{\boldsymbol{\theta}}(\cdot|c)}. \tag{28}$$

We can thus express the conditional squared MMD entirely in terms of Pauli-Z moments, which We estimate classically using the reference-state estimator in Sec. II A.

The bandwidth σ determines which correlation orders the objective emphasizes. The mask's Hamming weight $K = |a|$ counts the bits it selects, so it gives the correlation order of Z_a. Because we sample the mask coordinates independently,

$$\begin{aligned}K &\sim \text{Binomial}(n, p_\sigma),\\ \mathbb{E}[K] &= np_\sigma,\\ \text{Var}(K) &= np_\sigma(1-p_\sigma).\end{aligned} \tag{29}$$

Each bandwidth therefore induces a distribution over correlation orders. Large bandwidths favor lower-order correlations, whereas decreasing σ shifts weight toward higher-order correlations, with $\mathbb{E}[K] \to n/2$ as $\sigma \to 0$.

The bandwidth can also influence optimization behavior, as objectives dominated by global terms can exhibit barren plateaus [28–30]. Furthermore, if one tries to avoid them by emphasizing lower-order terms the loss reduces its sensitivity to higher-order differences between distributions [17], which may render the task unsolvable. Different strategies have been proposed to mitigate this issue, including tailored generator sets, restricted initializations and adversarial kernel adaptation [31–33]. However, determining how these approaches extend to our conditional framework remains beyond the scope of this work.

C. Unbiased finite-sample MMD and gradients

For a fixed condition c and mask a, we draw $B \geq 2$ independent target samples and compute

$$T_b = \chi_a(x_b), \qquad x_b \sim p(\cdot\mid c). \tag{30}$$

Independently of these target samples, we draw $S \geq 2$ independent reference samples and evaluate the reference-state estimator from Eq. (12),

$$Y_s = Y_{c,a}(z_s;\boldsymbol{\theta}), \qquad z_s \sim |\psi_c|^2. \tag{31}$$

We denote the corresponding sample means by

$$\overline{T} = \frac{1}{B}\sum_{b=1}^{B} T_b, \qquad \overline{Y} = \frac{1}{S}\sum_{s=1}^{S} Y_s. \tag{32}$$

We follow the independent-pair construction used for unbiased IQP MMD estimation [18, Appendix A.2.1], applying it to the conditional reference estimators and target moments.

The direct estimate $(\overline{Y}-\overline{T})^2$ overestimates the squared moment difference on average because it includes the sampling variances of both sample means. We remove this bias by excluding products that use the same draw twice:

$$\begin{aligned}\widehat{\Delta_c^2}(a) = &\frac{1}{S(S-1)}\sum_{s\neq t} Y_s Y_t - 2\overline{Y}\,\overline{T} \\ &+ \frac{1}{B(B-1)}\sum_{b\neq b'} T_b T_{b'}.\end{aligned} \tag{33}$$

Independence within each pair gives

$$\mathbb{E}\left[\widehat{\Delta_c^2}(a)\right] = \left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)^2. \tag{34}$$

Thus, Eq. (33) provides an unbiased finite-sample estimator of the squared character-moment difference appearing in Eq. (26). An unbiased finite-sample estimate of MMD^2 may be negative.

Once we have computed the sample values, we evaluate the estimator in time linear in the numbers of target and reference samples. Appendix A 2 gives the pair-sum identities used for this calculation.

The same construction gives an unbiased estimator of the phase gradients. For a phase parameter θ_j, let

$$Y'_s = \partial_{\theta_j} Y_{c,a}(z_s; \boldsymbol{\theta}). \tag{35}$$

We evaluate Y'_s using Eq. (14). Differentiating Eq. (33) gives

$$\partial_{\theta_j}\widehat{\Delta_c^2}(a) = \frac{2}{S(S-1)}\sum_{s\neq t} Y_s Y'_t - 2\overline{T}\,\overline{Y'}, \tag{36}$$

where

$$\overline{Y'} = \frac{1}{S}\sum_{s} Y'_s. \tag{37}$$

Taking the expectation yields

$$\mathbb{E}\left[\partial_{\theta_j}\widehat{\Delta_c^2}(a)\right] = 2\left(\mu_q(a \mid c) - \mu_p(a \mid c)\right)\partial_{\theta_j}\mu_q(a \mid c), \tag{38}$$

which is the gradient of the corresponding squared moment difference.

Since $|T_b| = 1$, independence of distinct draws and the triangle inequality in L^2 give

$$\begin{aligned}\mathbb{E}\left[\left(\widehat{\Delta_c^2}(a)\right)^2\right] &\leq 16, \\ \mathbb{E}\left[\left(\partial_{\theta_j}\widehat{\Delta_c^2}(a)\right)^2\right] &\leq 64.\end{aligned}$$

Averaging over conditions, bandwidths and sampled masks preserves these uniform bounds. Therefore, averaging N independent repetitions of the complete estimator gives mean-square error at most $16/N$ for the objective and $64/N$ for a specified phase-gradient component. By Chebyshev's inequality, $O(\epsilon^{-2})$ repetitions suffice for additive error ϵ at fixed confidence.

D. Stochastic mask estimator

Equation (26) expresses the squared MMD as an expectation over the mask distribution in Eq. (22). Since there are 2^n possible masks, we estimate this expectation using a finite sample of masks.

The identity mask $a = 0$ satisfies

$$\mu_p(0 \mid c) = \mu_q(0 \mid c) = 1 \tag{39}$$

for every condition and therefore contributes zero to the squared MMD. We consequently sample only nonidentity masks from the conditional distribution

$$P_\sigma(a \mid a \neq 0). \tag{40}$$

Under the original mask distribution, nonidentity masks have total probability

$$Q_\sigma^+ = 1 - (1-p_\sigma)^n. \tag{41}$$

We multiply the conditional expectation by Q_σ^+ to preserve the original Gaussian kernel objective.

We draw M nonidentity masks $a_1, \ldots, a_M$ independently from $P_\sigma(a \mid a \neq 0)$ and estimate the squared MMD at condition c as

$$\widehat{\mathrm{MMD}}_\sigma^2(c) = Q_\sigma^+ \frac{1}{M}\sum_{m=1}^{M}\widehat{\Delta_c^2}(a_m), \tag{42}$$

Here, $\widehat{\Delta_c^2}$ is the unbiased finite-sample estimator in Eq. (33). For each sampled condition and bandwidth, we use the same target and reference batches for all masks, drawing these batches independently of the masks. Sharing batches correlates the estimates across masks, but linearity of expectation preserves unbiasedness. Averaging over the mask, target and reference draws therefore recovers the full kernel squared MMD.

Repeated masks use the same target and reference batches, so they yield identical estimates and phase gradients. We therefore evaluate each distinct mask once and weight its contribution by its frequency among the M draws. This preserves the sampled objective and its gradient, as detailed in Appendix A 2.

IV. NUMERICAL EXPERIMENTS

In this section, we evaluate the empirical performance of conditional IQP models at the sixteen-qubit scale. We

also assess computational scalability by training models from 64 to 2048 qubits.

We train all models by minimizing the squared MMD objective. We use hyperparameter optimization for the two-coordinate, Curie–Weiss and scaling benchmarks. Appendices A and B describe the training settings, hyperparameter searches and evaluation protocols. Code and data are available in the accompanying repository [34].

For the 16-qubit experiments, our primary accuracy metric is the exact full-distribution total variation distance,

$$\mathrm{TVD}(p,q) = \frac{1}{2}\sum_{x\in\{0,1\}^n} |p(x) - q(x)|. \tag{43}$$

We also compare the Hamming-weight profiles using

$$\mathrm{TVD}_K(p,q) = \frac{1}{2}\sum_{k=0}^{n} |\rho_p(k) - \rho_q(k)|, \tag{44}$$

where $\rho_p(k)$ and $\rho_q(k)$ are the probabilities of observing Hamming weight k. This gives a coarser measure of agreement, with $\mathrm{TVD}_K(p,q) \le \mathrm{TVD}(p,q)$. For the larger models, we evaluate training progress using sampled squared MMD.

A. Conditional generation of binomial distributions

As an initial validation of the conditional generation framework, we consider a family of binomial distributions over Hamming weight. In this benchmark, the conditional IQP model must learn the dependence of the target distribution on an input condition $c \in [0,1]$, which smoothly controls its mean Hamming weight.

For $n = 16$ bits, we denote the Hamming weight by $K = |x|$ and define the target profile as

$$\rho(k \mid c) = \binom{16}{k} r(c)^k [1 - r(c)]^{16-k}, \tag{45}$$

where $k \in \{0, \ldots, 16\}$ and

$$r(c) = \frac{1}{1 + \exp[-\frac{3}{4}(2c-1)]}. \tag{46}$$

The target mean Hamming weight is $16r(c)$, which changes smoothly with c. Its variance is $16r(c)[1-r(c)]$, so varying the condition also changes the width of the profile. We assign equal probability to bitstrings with the same Hamming weight,

$$p(x \mid c) = \frac{\rho(|x| \mid c)}{\binom{16}{|x|}}. \tag{47}$$

We use a product-state encoder that represents the condition through input phases,

$$|\phi(c)\rangle = \bigotimes_{i=1}^{16} R_Z(c - 1/2) R_Y(\pi/2) \, |0\rangle_i \, . \tag{48}$$

This choice gives uniform computational-basis probabilities for every condition. The learned IQP unitary must therefore convert the encoded phases into the condition-dependent target probabilities. At $c = 0.5$, the encoder prepares $|+\rangle^{\otimes 16}$, which the IQP unitary preserves up to a global phase. The resulting uniform bitstring distribution exactly matches the target at this condition for every parameter vector. Agreement at the midpoint is therefore imposed by construction.

We parameterize the learned IQP unitary using a 4×4 lattice with sixteen one-body and twenty-four nearest-neighbor two-body generators, assigning an independent parameter to each generator. We train at $c \in \{0, 0.25, 0.5, 0.75, 1\}$ and evaluate at 80 held-out conditions distributed throughout $[0,1]$. We perform five independent fits. Appendix A 3 describes the training settings and checkpoint selection.

Figure 2 shows that the model reproduces the change in the target profiles and follows the target mean between training conditions. To quantify distributional accuracy, we average each fit's TVD over the 80 reporting conditions and take the median across fits. The full-distribution TVD is 0.0668, compared with 0.2848 for the encoder alone. The profile TVD is 0.0212, indicating closer agreement after grouping bitstrings by Hamming weight. Table I summarizes these errors and their variation across fits.

TABLE I. Exact full-distribution and profile TVDs for conditional generation of binomial distributions. Entries report the median across five independent fits of the mean error over eighty reporting conditions; brackets give the 25th–75th percentiles across fits. Lower is better.

Model	Full TVD	Profile TVD
Conditional IQP	0.0668 $[0.0624, 0.0692]$	0.0212 $[0.0171, 0.0219]$
Encoder only	0.2848	0.2848

These results show that the model learns the relation between a single input condition and the target distribution across the tested interval. We next consider a two-coordinate target family to test control of both mean and width, including combinations of conditions absent from training.

B. Two-coordinate conditioning and generalization

We now extend the binomial benchmark by introducing a second conditioning variable that changes the width of the target distribution beyond the constraint imposed by the binomial family. The aim is to test whether the conditional IQP model can learn the joint dependence on both variables and generate distributions at combinations absent from training. We focus on training conditions that vary one coordinate at a time, allowing us to evaluate generalization to joint conditions beyond the convex hull of the training set.

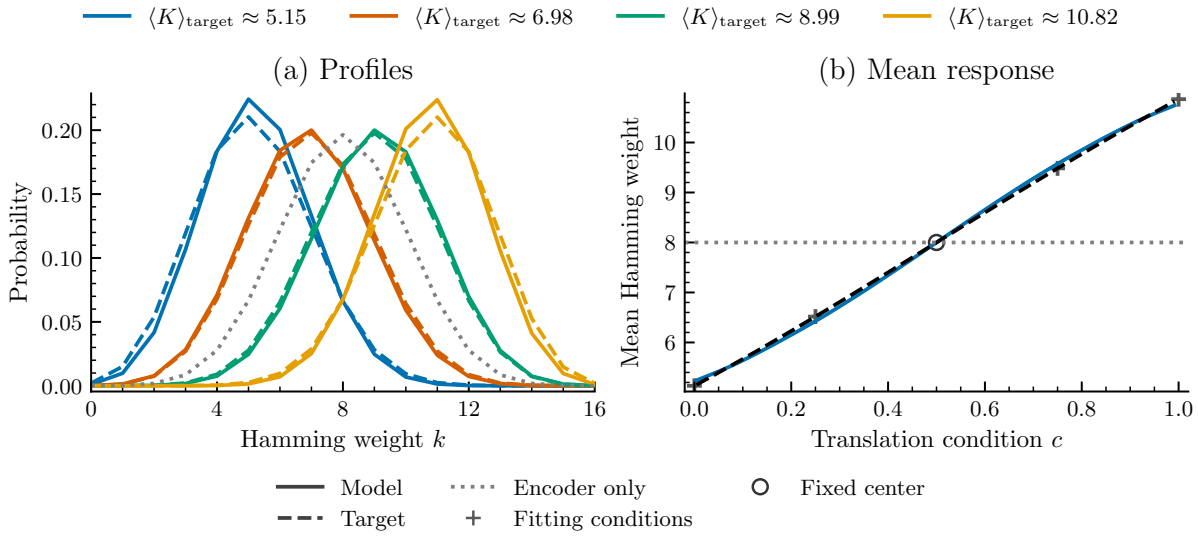


FIG. 2. Conditional generation of binomial distributions at sixteen qubits. (a) Solid colored curves show exact model Hamming-weight profiles averaged across five independent fits at four held-out conditions, $c \approx \{0.004, 0.329, 0.667, 0.992\}$. Dashed curves of the same color show the corresponding targets. The upper labels give the target mean Hamming weight for each color. (b) The solid blue curve and shading show the median and interquartile range of the model's mean Hamming weight across the five fits at 80 held-out conditions. The dashed black curve shows the target mean. Dotted gray curves show the encoder-only binomial profile and constant mean; gray crosses mark training conditions. The open circle at $c = 0.5$ marks exact agreement imposed by construction.

For conditions $u, v \in [-1, 1]$, we define the target Hamming-weight profile as

$$\rho_{u,v}(k) \propto \binom{16}{k} \exp\left[-\frac{(k-8-3u)^2}{8(1+0.2v)^2} + \frac{(k-8)^2}{8}\right], \quad (49)$$

with normalization over $k \in \{0, \ldots, 16\}$. As in the previous benchmark, we assign equal probability to bitstrings with the same Hamming weight,

$$p(x \mid u, v) = \frac{\rho_{u,v}(|x|)}{\binom{16}{|x|}}. \quad (50)$$

The coordinate u shifts the profile, while v changes its width. Both coordinates affect the actual mean and standard deviation, which are not given directly by $8 + 3u$ and $2(1 + 0.2v)$. We therefore evaluate the model against the mean and standard deviation computed from the target distribution.

Setting $v = 0$ and $u = 2c - 1$ recovers the binomial family from the previous benchmark. At $(u, v) = (0, 0)$, the target bitstring distribution is uniform, with a binomial Hamming-weight profile of mean 8 and standard deviation 2.

We extend the previous phase encoder by encoding u through single-qubit Z rotations and v through pairwise ZZ rotations. Defining $J_z = \frac{1}{2}\sum_{j=1}^{16} Z_j$, we prepare

$$|\phi(u, v)\rangle = e^{-i(u/2)J_z} e^{-i(v/40)J_z^2} |+\rangle^{\otimes 16}. \quad (51)$$

At $v = 0$, this recovers the product-state encoder of the previous benchmark under $u = 2c - 1$. The additional pairwise rotations generally produce correlated input states. Both operations change only the input phases, leaving the computational-basis probabilities uniform for every condition. The learned IQP unitary must therefore convert these phases into the varying target probabilities. At $(u, v) = (0, 0)$, the input is $|+\rangle^{\otimes 16}$, so agreement with the uniform target remains imposed by construction.

The encoded amplitudes depend only on Hamming weight, and the reference state $H^{\otimes 16} |\phi(u, v)\rangle$ retains this property. This structure allows efficient classical sampling and amplitude-ratio evaluation despite the input correlations. Appendix B describes these operations and the circuit preparation.

We retain the same IQP unitary architecture as in the binomial benchmark, with forty independent parameters on a 4×4 lattice. This unitary may not preserve the permutation symmetry of the target distribution, so we evaluate accuracy using full-distribution TVD as well as Hamming-weight profile TVD.

We train the model using conditions on the two coordinate axes, which we refer to as axis training. These conditions have the form $(u, 0)$ or $(0, v)$, so only one coordinate varies at a time. We use nine distinct training conditions, including the origin. Their convex hull is the diamond-shaped region $|u| + |v| \leq 1$.

We select hyperparameters and checkpoints using additional conditions restricted to the same axes. Thus, neither training nor model selection uses mixed conditions, defined here as pairs with both $u \neq 0$ and $v \neq 0$. We evaluate the selected models at sixteen mixed pairs.

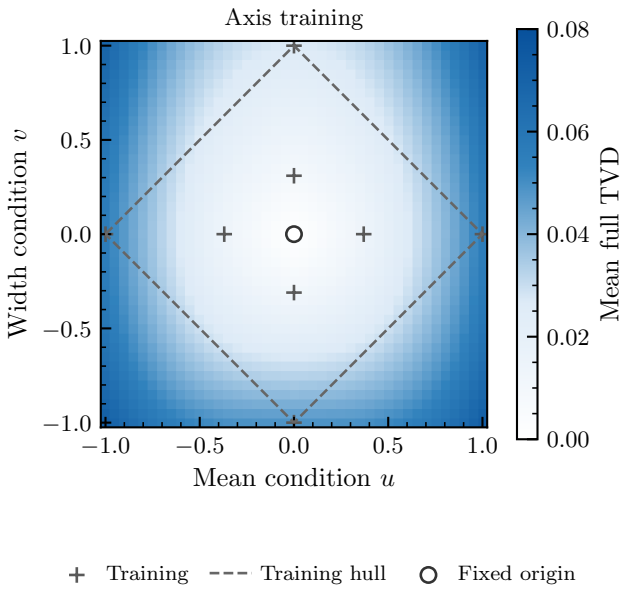


FIG. 3. Joint conditional generation after axis-only training. Colors show the mean full-distribution TVD across five independent fits on a 41×41 diagnostic grid. Gray crosses mark training conditions, and the dashed diamond marks their convex hull. The open circle at $(u, v) = (0, 0)$ marks the uniform target and model distribution fixed by construction. The diagnostic grid includes training and development conditions; quantitative reporting uses sixteen separate mixed conditions excluded from training and model selection.

Each individual coordinate value appears in the training set, but their combinations do not. Four evaluation pairs lie inside the training convex hull and twelve lie outside it. This tests the learned joint response at combinations absent from training and model selection, while remaining within the observed range of each coordinate.

We perform five independent final fits. Selection uses exact full-distribution TVD at the axis-only development conditions. Appendix B specifies the search and evaluation protocol and reports a complementary grid-training experiment.

After axis training, the model approximates the target distributions at joint conditions excluded from both training and model selection. We first average full-distribution TVD over the evaluation conditions within each fit, then report the mean and sample standard deviation across five fits. The sixteen mixed pairs give a TVD of 0.0468 ± 0.0143. The four pairs inside the training convex hull give 0.0185 ± 0.0060, while the twelve outside give 0.0563 ± 0.0171. The model therefore approximates joint target distributions beyond the training convex hull. The higher error outside the hull does not by itself isolate an extrapolation penalty, because the target conditions also differ between the two subsets.

The mean and width responses clarify what the models learn. Figure 10 shows that the model mean closely follows the target across the tested widths. Figure 9 shows that the model also captures the increase in width with v, although it tends to overestimate the narrowest target distributions. These results support joint conditional control of the mean and width, including at combinations absent from training.

This experiment tests a shared transformation across a smooth, task-informed family. It demonstrates responses at combinations absent from fitting and development selection, but do not establish that the lattice is necessary or that the selected encoder generalizes to unrelated families. The condition sets and phase features were explored during earlier design, and exact development targets contribute to selection. We therefore interpret the results as conditional capability within the studied domain.

C. Curie–Weiss spin model

We next test whether conditional IQP models can learn a temperature-dependent change in distribution shape. We consider the zero-field Curie–Weiss spin model [35], whose magnetization profile changes from a single central peak to two symmetric peaks as temperature decreases.

For $n = 16$ spins, we represent each configuration by a bitstring $x \in \{0, 1\}^n$ and assign $s_i = 2x_i - 1$. The magnetization per spin is therefore $m(x) = n^{-1} \sum_i s_i = (2|x| - n)/n$. With unit interaction strength and inverse temperature β, the target distribution is

$$p(x \mid \beta) = \frac{1}{\mathcal{Z}_\beta} \exp\left[\frac{\beta}{2n}(2|x| - n)^2\right], \tag{52}$$

where $\mathcal{Z}_\beta$ normalizes the distribution. The target assigns equal probability to bitstrings with the same Hamming weight and is invariant under global spin flip.

We use a parity encoder to encode inverse temperature while preserving the target's global spin-flip symmetry. The input state is

$$|\phi_{\mathrm{P}}(\beta; s)\rangle = \exp\left[-\frac{is\beta}{2} \sum_{i=1}^{15} Z_i Z_{16}\right] |+\rangle^{\otimes 16}, \tag{53}$$

where s is an encoder scale selected by hyperparameter optimization. These pairwise rotations introduce temperature-dependent phase correlations while leaving the computational-basis probabilities uniform. The shared IQP unitary must therefore convert the encoded phases into the target probabilities.

The name refers to the reference state $|\psi_{\mathrm{P}}\rangle = H^{\otimes 16} |\phi_{\mathrm{P}}\rangle$, which has support only on bitstrings with even parity. Consequently, the encoded input satisfies $X^{\otimes 16} |\phi_{\mathrm{P}}\rangle = |\phi_{\mathrm{P}}\rangle$. Since the learned IQP unitary commutes with $X^{\otimes 16}$, the model satisfies

$$q_{\boldsymbol{\theta}}(x \mid \beta) = q_{\boldsymbol{\theta}}(\bar{x} \mid \beta) \tag{54}$$

for every parameter vector, where $\bar{x}$ flips all bits. This enforces zero signed mean magnetization and a symmetric

magnetization profile. It does not impose permutation symmetry or agreement with the target probabilities.

The reference state supports classical sampling and amplitude-ratio evaluation in $O(n)$ arithmetic operations for the corresponding n-qubit construction. Appendix A 4 gives the preparation circuit and reference amplitudes. The encoder requires no ancillary qubits.

We parameterize each IQP unitary with all sixteen one-body and 120 two-body generators, giving 136 independent parameters per circuit. We train the conditional IQP model at $\beta \in \{0.1, 0.7, 1.3, 1.95\}$ and evaluate it at $\beta \in \{0.4, 1, 2\}$ and on a separate dense grid. The three evaluation temperatures are excluded from both conditional IQP training and model selection.

We compare the conditional IQP model with two families of independently trained circuits. Each family contains three circuits, one trained directly at each evaluation temperature. The conditional IQP model and temperature-specific IQP models use the parity encoder, whereas the temperature-specific models with standard input use $|0\rangle^{\otimes 16}$. Within each baseline family, the circuits have separate learned parameters but use one jointly selected hyperparameter configuration. This comparison tests conditional generation at unseen temperatures against models trained specifically for those targets.

Appendix A 4 b also compares the conditional IQP model with a shared product-state encoder using the same IQP ansatz. Replacing the encoder changes both the input correlations and the enforced spin-flip symmetry, so this comparison assesses the encoder construction as a whole.

We train by minimizing sampled squared MMD and optimize hyperparameters separately for each model family. For checkpoint and final hyperparameter selection, we use exact probabilities to compute a full-distribution TVD-based score. Among candidates within a fixed tolerance of the best score, we select the candidate with the lowest combined error score for mean absolute magnetization and energy fluctuations. Appendix A 4 a defines these scores and specifies the optimization budgets and selection protocol. We perform ten independent final repetitions for each family and report means and sample standard deviations across all ten.

Figure 4 shows that the conditional IQP model captures the change from a central magnetization peak at $\beta = 0.4$ to two peaks at large absolute magnetization at $\beta = 2$. Shape discrepancies remain near $\beta = 1$. The conditions $\beta = 0.4$ and 1 test interpolation, while $\beta = 2$ tests slight extrapolation beyond the largest training inverse temperature, 1.95. These conditions are excluded from training and model selection, although earlier development explored this temperature range.

On the separate 101-point evaluation grid in $[0, 2]$, we compute each repetition's inverse-temperature-averaged full-distribution TVD and maximum TVD. Their means and sample standard deviations across ten repetitions are 0.1033 ± 0.0036 and 0.1406 ± 0.0037, respectively. The error averaged across repetitions peaks near $\beta = 0.98$, placing the largest discrepancies near the crossover.

Table II compares the models at the three evaluation temperatures. Conditional IQP has lower mean full-distribution TVD than temperature-specific IQP at $\beta = 1$ and 2, while the temperature-specific model performs better at $\beta = 0.4$. At $\beta = 2$, their TVDs are 0.0832 ± 0.0103 and 0.1525 ± 0.0021, respectively. At $\beta = 1$, the conditional model has lower full-distribution error despite larger profile error. The standard-input baseline has the largest mean full-distribution TVD at all three temperatures.

At $\beta = 0$, the encoder prepares $|+\rangle^{\otimes 16}$, which the IQP unitary changes only by a global phase. Exact agreement with the uniform target at this endpoint therefore holds for every parameter vector.

We also examine the temperature dependence of mean absolute magnetization and energy fluctuations. We use $\langle |m| \rangle$ because the parity encoder enforces zero signed mean magnetization. For the energy $H = -nm^2/2$, we compute the per-spin energy-fluctuation statistic

$$c_{\mathrm{fluc}}(\beta) = \frac{\beta^2}{n}\big(\langle H^2 \rangle - \langle H \rangle^2\big). \tag{55}$$

For the Gibbs target, this expression gives the equilibrium heat capacity per spin. For learned distributions, we use it as an energy-fluctuation diagnostic. Appendix A 4 d details the calculations.

Figure 5 shows that conditional IQP captures the increase in mean absolute magnetization and the peaked energy-fluctuation response as inverse temperature increases. Discrepancies remain in both observables. In the averaged response, the fluctuation peak is lower and occurs at a larger inverse temperature than the target peak.

Observable accuracy does not follow the full-distribution TVD ranking. At $\beta = 1$, temperature-specific IQP reproduces mean absolute magnetization more accurately, while conditional IQP better matches energy fluctuations. At $\beta = 2$, conditional IQP has lower full-distribution TVD but larger fluctuation error. The standard-input baseline also shows substantial variation across repetitions at this condition. Table IX reports the observable errors.

D. Scaling

We next assess the computational feasibility of classical conditional IQP training at larger system sizes. We train Curie–Weiss models with $n \in \{64, 128, 256, 512, 1024, 2048\}$ qubits using the parity encoder introduced above, without ancillas. This encoder enforces global spin-flip symmetry at every size. We assign independent trainable phases to each qubit and each horizontal or vertical nearest-neighbor pair on an open rectangular lattice. An $L_x \times L_y$ lattice therefore has $3n - L_x - L_y$ trainable parameters.

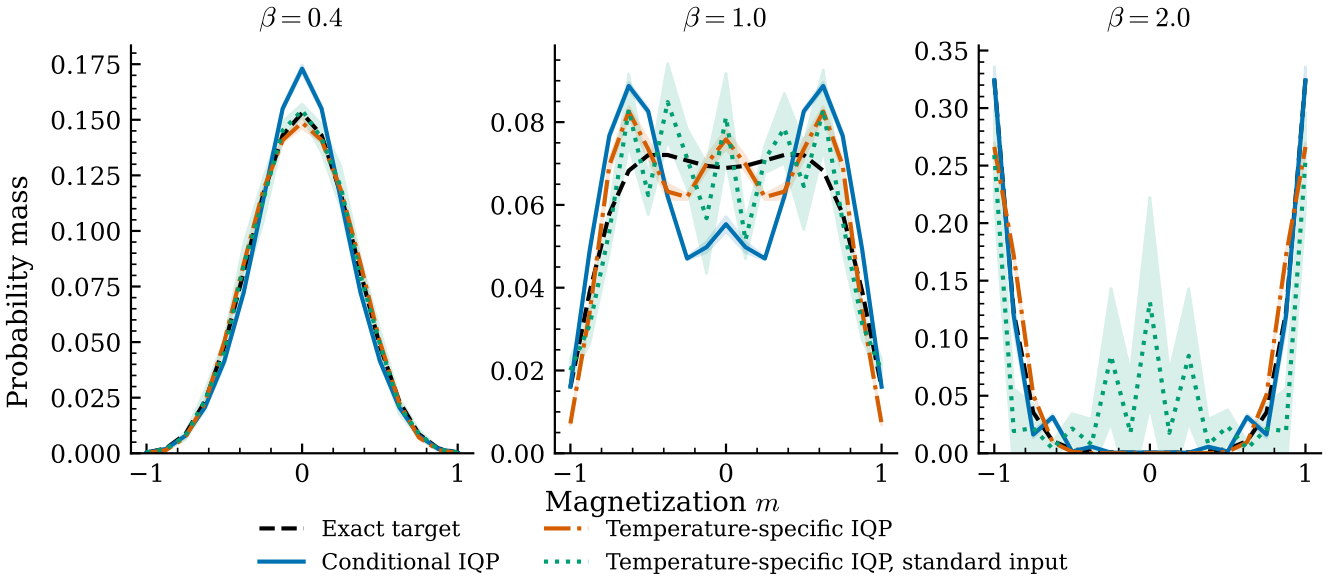


FIG. 4. Curie–Weiss magnetization profiles at sixteen qubits. Magnetization per spin is $m = (2k - 16)/16$, where k is Hamming weight. Colored curves show exact probability masses averaged across ten independent repetitions; shading shows one sample standard deviation above and below the mean. Dashed black curves show the targets. Conditional IQP and temperature-specific IQP use the parity encoder; the standard-input baseline uses $|0\rangle^{\otimes 16}$. The displayed temperatures are excluded from conditional training and model selection, while the temperature-specific models train directly at them. Panels use separate vertical scales. Table II reports full-distribution errors.

TABLE II. Exact full-distribution TVD at sixteen qubits. Values are means ± sample standard deviations over ten fresh final repeats, computed within each repeat before aggregation. Lower is better. Conditional IQP uses the shared parity encoder and excludes markers from fitting and selection; the temperature-specific IQP models use the parity encoder or the standard input $|0\rangle^{\otimes 16}$ and fit directly at each marker.

Model	$\beta = 0.4$	$\beta = 1$	$\beta = 2$
Conditional IQP	0.1185 ± 0.0107	0.1396 ± 0.0048	0.0832 ± 0.0103
Temperature-specific IQP	0.0692 ± 0.0059	0.1477 ± 0.0081	0.1525 ± 0.0021
Temperature-specific IQP, standard input	0.3787 ± 0.0034	0.3400 ± 0.0074	0.4101 ± 0.1629

We optimize hyperparameters at 64 and 256 qubits and use the selected configuration at all six sizes. We train at inverse temperatures $\beta \in \{0.6, 0.8, 1.2, 1.4\}$ and evaluate at held-out values $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$, which do not enter checkpoint or hyperparameter selection. At each size, we perform three runs with three initializations per run. Validation at the training conditions selects a checkpoint and its associated initialization within each run. We report the full training trajectory of that initialization. Appendix A 5 specifies the optimization and evaluation protocols.

We use two kernel bandwidths, chosen to give expected mask weights of two and four under the mask distribution in Eq. (22). This choice emphasizes low-order Pauli-Z moments at every system size. The corresponding bandwidths therefore increase with n. We assess performance using classical estimates of the squared MMD, averaged over the held-out inverse temperatures and both bandwidths.

Figure 6 shows that the held-out MMD^2 decreases at every size before reaching a plateau. Over 5000 updates, the median decreases from 0.0419 to 0.00379 at 64 qubits and from 0.0420 to 0.00708 at 2048 qubits. These results demonstrate the feasibility of classical conditional training up to 2048 qubits and improvement at conditions excluded from training and selection.

The decrease in MMD^2 does not by itself establish accurate recovery of the full target distribution. Although the kernels retain all correlation orders, they emphasize low-order moments, so higher-order discrepancies may remain weakly penalized or unresolved by finite sampling. The size-dependent bandwidths also limit direct comparisons of distribution accuracy across system sizes.

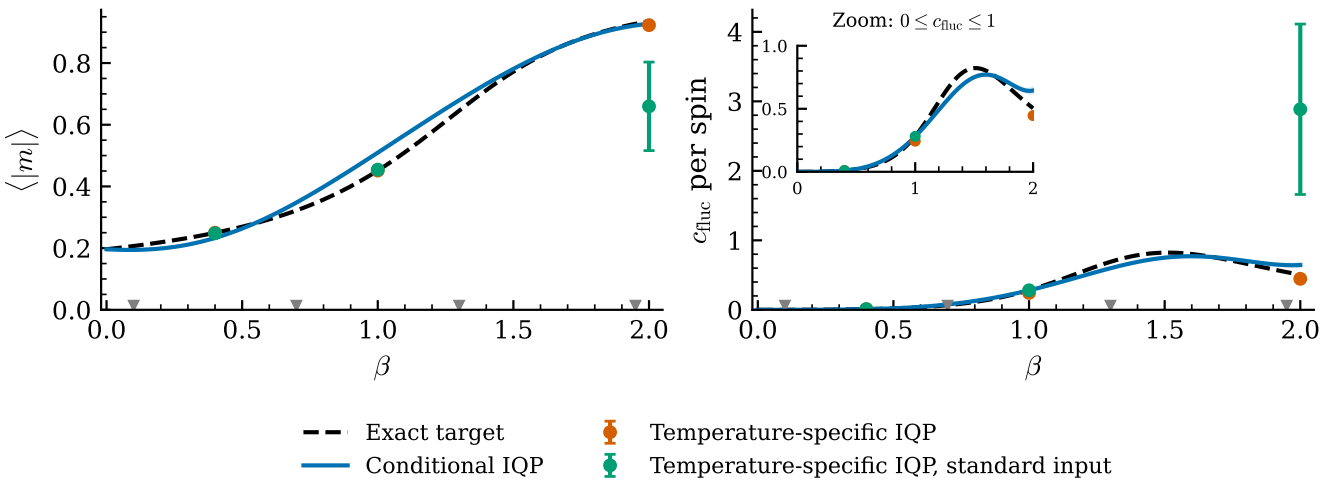


FIG. 5. Mean absolute magnetization and per-spin energy fluctuations for sixteen-qubit Curie–Weiss models. Dashed black curves show the targets. Blue curves and shading show conditional IQP means and one sample standard deviation across ten independent repetitions. Temperature-specific models appear as marker means with sample-standard-deviation error bars at their training conditions, $\beta = 0.4, 1, 2$. Gray triangles mark the four conditional training conditions. The inset enlarges the fluctuation response over $0 \le c_{\mathrm{fluc}} \le 1$, while the main panel retains the full range. We compute each observable within each repetition before aggregation. Equation (55) defines the fluctuation statistic.

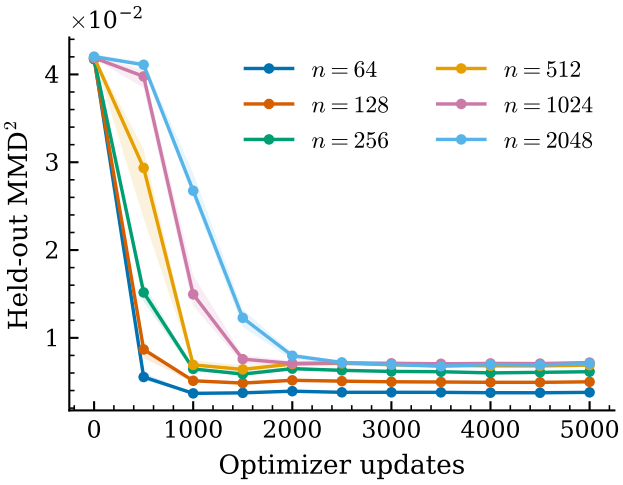


FIG. 6. Classical training of conditional IQP Born machines with 64 to 2048 qubits. All models use the parity encoder and an open rectangular lattice of trainable one- and two-body phases. At each plotted update, we classically estimate MMD^2 and average over held-out inverse temperatures $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$ and two kernel bandwidths. The bandwidths give expected mask weights of two and four and vary with system size. Each run contributes the trajectory of the initialization selected by validation at the training conditions. Curves and shaded bands show the median and interquartile range across three runs per size.

V. INPUT-STATE INDUCED SYMMETRIES IN IQP BORN MACHINES

The Curie–Weiss encoder enforces global spin-flip symmetry through the input state. We now extend this construction to several prescribed bit-flip symmetries while retaining efficient classical sampling and amplitude-ratio evaluation. For a bit mask $h \in \{0,1\}^n$, the operator $X_h = \bigotimes_{i=1}^n X_i^{h_i}$ commutes with the IQP unitary. If the encoded input is an eigenstate of X_h, applying the unitary preserves its eigenvalue. The output amplitudes at x and $x \oplus h$ then differ at most by a sign, so their probabilities are equal. This principle underlies the input-state construction of Recio-Armengol *et al.* [18, Sec. 6]. Such symmetry constraints provide inductive biases in variational quantum models [36].

We construct reference states whose computational-basis configurations satisfy prescribed parity constraints. Let $h_1, \ldots, h_r \in \mathbb{F}_2^n$ be linearly independent masks specifying the desired bit-flip symmetries. Choose a matrix $C \in \mathbb{F}_2^{n \times k}$, with $k = n - r$, whose columns form a basis of the subspace orthogonal to all these masks. For any string of latent bits $u \in \mathbb{F}_2^k$, the reference configuration Cu therefore satisfies $h_j^\top Cu = 0$ for every j. All matrix operations below are over $\mathbb{F}_2$.

For each condition c and latent bit $\ell = 1, \ldots, k$, choose efficiently evaluable complex amplitudes $f_\ell(0; c)$ and $f_\ell(1; c)$ satisfying

$$|f_\ell(0; c)|^2 + |f_\ell(1; c)|^2 = 1. \tag{56}$$

We define the reference state and encoded input as

$$|\psi_c\rangle = \sum_{u\in\mathbb{F}_2^k}\left[\prod_{\ell=1}^{k} f_\ell(u_\ell;c)\right]|Cu\rangle\,, \qquad |\phi(c)\rangle = H^{\otimes n}\,|\psi_c\rangle\,. \tag{57}$$

The matrix C fixes the parity constraints, while the factor amplitudes encode the condition. Since C is injective, distinct latent strings give distinct reference configurations, and the factor normalization ensures that the reference state is normalized.

To sample the reference distribution, we draw each latent bit independently with probability $|f_\ell(1;c)|^2$ of obtaining one and return Cu. If a mask a belongs to $\operatorname{im} C$, there is a unique latent displacement $d \in \mathbb{F}_2^k$ satisfying $a = Cd$. Flipping the reference bits selected by a then maps Cu to $C(u \oplus d)$, giving

$$R_c(Cu,a) = \prod_{\ell=1}^{k}\frac{f_\ell(u_\ell\oplus d_\ell;c)}{f_\ell(u_\ell;c)}. \tag{58}$$

If $a \notin \operatorname{im} C$, the flipped configuration lies outside the reference support and the ratio is zero. We evaluate denominators only at sampled configurations with nonzero amplitude.

The space $\ker C^\top$ is the span of the prescribed symmetry masks. For every h in this space, the parity constraints imply $Z_h\,|\psi_c\rangle = |\psi_c\rangle$, and the Hadamard layer gives $X_h\,|\phi(c)\rangle = |\phi(c)\rangle$. Since $U_{\boldsymbol{\theta}}$ commutes with X_h, the output retains this eigenvalue. Consequently,

$$q_{\boldsymbol{\theta}}(x \mid c) = q_{\boldsymbol{\theta}}(x\oplus h \mid c) \qquad \text{for all } h \in \ker C^\top. \tag{59}$$

The construction therefore guarantees a group of 2^r bit-flip symmetry transformations for every condition and parameter vector. Particular factor amplitudes or parameter values may produce additional symmetries.

The Curie–Weiss encoder is a special case with $Cu = (u, \bigoplus_\ell u_\ell)$. Its reference configurations have even parity, which gives global spin-flip symmetry after the Hadamard layer. More generally, the construction supports reference sampling and amplitude-ratio evaluation at polynomial cost, provided that the factor amplitudes admit efficient evaluation with sufficient precision. Appendix C establishes normalization, access costs and state preparation. The following subsection illustrates how this construction enforces several independent symmetries in a signed-graph model.

A. Signed-graph encoding with exact switching symmetry

We illustrate the construction with a sixteen-qubit encoder that enforces six independent bit-flip symmetries. We consider a temperature-dependent family of signed-graph distributions whose triangle interactions remain invariant under vertex switches. This example demonstrates encoder design and classical access; we do not evaluate learning accuracy for this family.

Sixteen qubits represent the edge signs of a seven-vertex graph. We connect every pair among vertices $0,\dots,4$, and connect each of vertices 5 and 6 to vertices $0,1,2$, as shown in Figure 7(a). Each edge bit x_{ij} specifies a sign $s_{ij} = (-1)^{x_{ij}}$. We define the Gibbs family

$$p_\beta(x) = Z_\beta^{-1}e^{\beta S(x)}, \qquad S(x) = \sum_{\{i,j,k\}\in\mathcal{T}} s_{ij}s_{ik}s_{jk}, \tag{60}$$

where β is the inverse temperature, Z_β normalizes the distribution, and $\mathcal{T}$ contains the graph's sixteen triangles. The triangle-balance score $S(x)$ assigns a contribution of $+1$ to each balanced triangle, whose edge signs have positive product, and -1 otherwise. Increasing β therefore favors configurations with more balanced triangles.

A vertex switch reverses the signs of all edges incident on that vertex. Every triangle contains either zero or two of these edges, so the switch preserves its edge-sign product and hence p_β. To describe combinations of switches, let $\eta_i \in \{-1,+1\}$ indicate whether vertex i is switched. The combined action replaces s_{ij} by $\eta_i s_{ij}\eta_j$. Switching all seven vertices leaves every edge unchanged, so we fix $\eta_0 = 1$ to remove duplicate representations. The remaining six choices generate 64 distinct transformations, including the identity. We denote their edge-bit masks by G, so a switch acts as $x \mapsto x \oplus g$ for $g \in G$.

To construct the encoder, we use the six root edges $0i$, with $i = 1,\dots,6$, as a spanning tree. We call the remaining ten edges chords and denote their set by $\mathcal{C}$. Each chord ij forms a triangle with root edges $0i$ and $0j$. These ten triangles form a cycle basis. In the preceding construction, each column of C is the edge-bit mask of one such triangle, and each latent bit determines whether to include that triangle in the reference configuration. Combining the selected triangles modulo two gives even incidence parity at every vertex.

We prepare this reference state by initializing all sixteen qubits in $|0\rangle$. For each chord ij, we apply $R_X(t_\beta)$ to its qubit and two CNOT gates controlled by that qubit, targeting root edges $0i$ and $0j$. On an isolated triangle, these operations prepare $\cos(t_\beta/2)\,|000\rangle - i\sin(t_\beta/2)\,|111\rangle$, as illustrated in Figure 7(b). In the full graph, the CNOTs combine contributions to shared root edges modulo two. After processing all ten chords, we apply one Hadamard layer to all sixteen qubits to obtain the encoded input.

The encoded input has the form

$$|\phi_\beta\rangle = \exp\left[-\frac{it_\beta}{2}\sum_{(i,j)\in\mathcal{C}} Z_{0i}Z_{0j}Z_{ij}\right]|+\rangle^{\otimes 16}, \tag{61}$$

$$t_\beta = \frac{\pi}{2}\tanh(1.5\beta). \tag{62}$$

We choose this temperature map as a concrete example; the symmetry guarantee does not depend on its form.

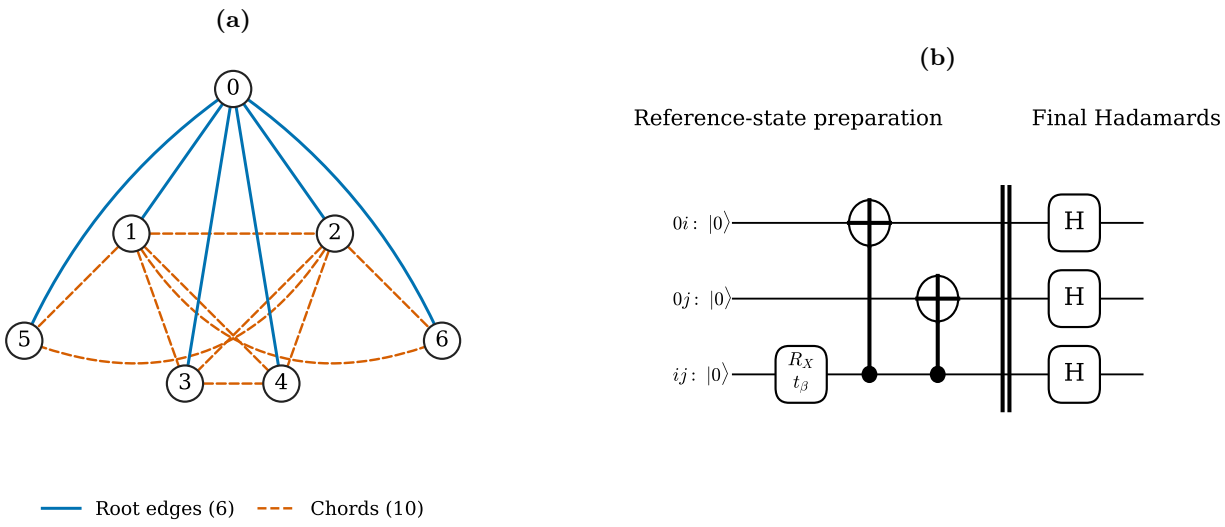


FIG. 7. Signed-graph encoder with six independent switching symmetries. (a) The seven-vertex graph has sixteen edges. Six root edges form a spanning tree, and each of the ten remaining edges forms a fundamental triangle with two root edges. (b) Reference-state preparation on an isolated triangle. A chord rotation $R_X(t_\beta)$ and two CNOTs prepare $\cos(t_\beta/2)\,|000\rangle - i\sin(t_\beta/2)\,|111\rangle$ before the displayed Hadamards. For the full encoder, we repeat the rotation–CNOT construction over all ten chords, combining contributions to shared root edges modulo two, then apply one global Hadamard layer. All rotations use $t_\beta = (\pi/2)\tanh(1.5\beta)$.

The diagonal operators leave the computational-basis probabilities uniform, so temperature enters through correlated phases. The shared IQP unitary can convert these phases into condition-dependent output probabilities.

The preparation uses ten rotations, twenty CNOTs and sixteen Hadamards, without ancillary qubits. The state before the Hadamard layer is the reference state used for classical estimation. To sample it, we draw each chord bit independently with probability $\sin^2(t_\beta/2)$ of obtaining one. Each root-edge bit is then the sum, modulo two, of the chord bits incident on its nonzero vertex. The chord bits also determine each supported amplitude through a product of ten rotation coefficients, giving direct amplitude-ratio access. Appendix D provides the explicit expressions.

Even incidence parity implies that every supported reference configuration has even parity on the edges incident on any switched vertex. The Hadamard layer therefore gives

$$X_g\,|\phi_\beta\rangle = |\phi_\beta\rangle\,, \qquad g \in G, \tag{63}$$

where X_g flips the edge bits selected by g. Since the IQP unitary commutes with these operators, the model satisfies

$$q_{\boldsymbol{\theta}}(x \mid \beta) = q_{\boldsymbol{\theta}}(x \oplus g \mid \beta), \qquad g \in G, \tag{64}$$

for every condition and parameter vector. Thus, the encoder enforces all six independent switching symmetries while retaining classical sampling and amplitude-ratio access, without restricting the learned IQP phases.

VI. DISCUSSION

Our framework separates condition encoding from the shared unitary learned across a family of distributions. Classical estimation of the training objective and its gradients requires efficient sampling and amplitude-ratio evaluation for the reference state, rather than an encoder restricted to IQP form. This distinction allows us to use input-state design both to encode conditions and to impose exact output symmetries while retaining classical training.

The binomial benchmark validates learning the dependence of a target distribution on a single condition, while its two-coordinate extension tests learning a joint conditional response. In particular, axis training exposes the model to each coordinate separately, yet the model approximates target distributions at combinations excluded from training and model selection, including outside the training convex hull. Errors are larger outside this hull, although this comparison does not isolate the effect of extrapolation from variation in target difficulty. These results support conditional generation within the studied family, using a task-informed encoder and exact development distributions for model selection; they do not establish generalization to unrelated distribution families.

The Curie–Weiss experiment distinguishes the conditional response learned by the model from the symmetry imposed by its encoder. The shared IQP unitary captures the temperature-dependent change from a central magnetization peak to two separated peaks, including at tem-

peratures excluded from training and model selection. The parity encoder guarantees global spin-flip symmetry for every parameter vector, but this guarantee alone does not ensure accurate probabilities or physical observables. The parity encoder gives lower full-distribution errors than the product encoder under the tested protocol, although this comparison changes both input correlations and phase generators and does not isolate the effect of symmetry. Comparisons with temperature-specific IQP models depend on temperature and metric; neither approach is consistently more accurate. The temperature-specific models use separate parameter sets and a larger total optimizer-update budget in this comparison.

The signed-graph construction shows that condition encoding and efficient classical reference access can coexist with several prescribed output symmetries. In this example, the encoder enforces six independent vertex-switching symmetries for every condition and parameter vector while leaving the learned IQP phases unrestricted. The guarantee follows from input-state design and does not depend on training accuracy. This construction establishes symmetry control and classical access, but does not demonstrate learning accuracy for the associated Gibbs family.

Efficient, unbiased estimation of the MMD objective does not guarantee accurate recovery of the full target distribution [37]. Although the Gaussian kernel includes all correlation orders, its bandwidth determines their relative weights, and finite sampling and optimization may leave weakly weighted discrepancies unresolved. Hamming-weight profiles also provide only a coarse measure of agreement, since they group bitstrings that may have different model probabilities. For example, at $\beta = 0.4$, the temperature-specific IQP model with standard input has mean profile TVD 0.0281 but mean full-distribution TVD 0.3787. Full-distribution evaluation is therefore important whenever feasible.

The Curie–Weiss reporting temperatures are excluded from training and model selection. However, earlier model development explored the temperature range, and selection uses exact target distributions and a physical-observable criterion. The results therefore assess generalization within an already studied family and range. Variation across ten independent fits characterizes training variability, while evaluation at finitely many temperatures does not establish accuracy throughout the continuous interval.

The larger models demonstrate classical conditional training up to 2048 qubits and reductions in classically estimated MMD^2 at held-out conditions. These models use sparse lattice connectivity, whereas the sixteen-qubit Curie–Weiss models include all two-body interactions. The scaling results therefore concern a different circuit architecture. Moreover, the kernels emphasize low-order moments, so the observed objective reductions do not establish full-distribution accuracy. Their size-dependent bandwidths also limit direct comparisons of accuracy across system sizes.

Future work should examine how encoder design affects conditional accuracy and the costs of reference-state estimation and quantum state preparation. Broader families of correlated reference states and multidimensional encodings could extend the tasks addressed by this framework. Optimization also requires further study, particularly whether gradients from different conditions reinforce or oppose one another. Generator selection, initialization and adaptive kernels provide directions for investigating these questions [31–33].

Quantum deployment raises additional questions about how noise and circuit partitioning affect conditional accuracy and the imposed output symmetries [38, 39]. The binomial, two-coordinate and Curie–Weiss target families studied here admit efficient classical sampling, so these experiments do not demonstrate a quantum sampling advantage. Assessing such an advantage would require comparisons with strong classical sampling methods, together with sampling-hardness and verification arguments appropriate to the trained conditional circuits.

VII. CONCLUSIONS

We introduced a conditional formulation of IQP Born machines that jointly trains a shared IQP unitary across a family of distributions. Efficient sampling and amplitude-ratio evaluation for condition-dependent reference states yield unbiased classical estimates of the conditional MMD objective and its gradients. The encoder need not have IQP form, and input-state design can enforce prescribed output bit-flip symmetries for every condition and learned parameter vector.

At sixteen qubits, the binomial benchmark and its two-coordinate extension demonstrate conditional generation, including at joint conditions outside the training convex hull. The Curie–Weiss experiment shows that a shared model can capture the temperature-dependent change from unimodal to bimodal magnetization profiles. The signed-graph construction illustrates how to impose several independent output symmetries while retaining efficient classical reference access. Training runs up to 2048 qubits demonstrate computational feasibility and reductions in classically estimated MMD^2 at held-out conditions.

These results extend classically trainable IQP Born machines to conditional generation and input-induced symmetry control. Full-distribution accuracy at larger scales and practical quantum sampling advantage remain open.

DATA AVAILABILITY

The code and data supporting this study are available on GitHub [34].

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Appendix A: Hyperparameters and evaluation protocols

1. Common training settings

We train all models using the unbiased estimator of the squared-MMD objective and keep the encoders fixed within each fit. At each update, we draw one training condition and one kernel bandwidth uniformly, then sample a target minibatch with replacement. We draw fresh nonidentity masks and apply the probability correction for excluding the identity mask. Mask budgets count draws, including repeated masks. We draw reference samples independently of the target minibatch and reuse them across masks.

Each initialization starts from a new circuit parameter vector and resets the optimizer while retaining the target pools. For the binomial benchmark and scaling experiments, we draw the initial phases independently as $\theta_j \sim \mathcal{N}(0, s_\theta^2)$. A training run comprises the prescribed initializations. For these two experiments, we select the checkpoint with the lowest selection score across all initializations within each run. Reported uncertainty summarizes variation across runs.

We train the binomial and scaling models with `Adam` [40] and cosine learning-rate decay. Table III gives their training budgets, initialization widths, initial learning rates, final-to-initial learning-rate ratios and moment coefficients. Both experiments use $\epsilon = 10^{-8}$ and zero weight decay. Section A 4 a specifies the Curie–Weiss hyperparameter search and selection protocol, with selected settings in Table VI. Appendix B describes the two-coordinate protocol.

TABLE III. Training settings for the binomial benchmark and final scaling runs. Mask and reference budgets count draws per update. The Curie–Weiss experiments use the separate budgets in Table IV. Tuned scaling values are rounded to six significant digits.

Setting	Binomial benchmark	Scaling
Updates per initialization	1,024	5,000
Initializations per run	1	3
Reported runs	5	3
Training targets per condition	512	10,000
Target minibatch	128	450
Training mask draws	1,024	512
Training reference draws	1,024	1,024
Expected mask weights	$\{1\}$	$\{2, 4\}$
Checkpoint interval	32	100
Optimizer	`Adam`	`Adam`
Learning rate	0.005	0.00189326
Learning-rate schedule	Cosine	Cosine
Initialization width s_θ	0.25	0.0129803
Coefficient b_1	0.9	0.822126
Final-to-initial learning-rate ratio	0.1	0.119851
Coefficient b_2	0.999	0.962419

For an expected mask weight w, we set the Gaussian bandwidth to

$$\sigma(n, w) = \left[-2 \ln\left(1 - \frac{2w}{n}\right)\right]^{-1/2}, \qquad 0 < w < n/2, \tag{A1}$$

where n is the number of visible qubits and w is the expected mask weight before excluding the identity mask. We weight the bandwidth entries equally in the objective and sample an entry uniformly at each update. The sixteen-qubit Curie–Weiss experiments use three entries, two of which we select by hyperparameter optimization as specified in Table V. Repeated entries each retain their sampling weight, so repeating a bandwidth increases its contribution to the objective. The upper grid value 7.99 keeps $w < n/2$ and the bandwidth strictly positive.

We report rotation angles in radians and compute sixteen-qubit full-distribution and Hamming-weight profile TVDs from exact circuit probabilities. These TVD calculations do not use the sampled MMD estimates.

2. Efficient evaluation of the sampled estimator

For the sample values defined in Sec. III C, we compute the pair sums using

$$\sum_{s\neq t} Y_s Y_t = \left(\sum_s Y_s\right)^2 - \sum_s Y_s^2, \tag{A2}$$

and, since $T_b^2 = 1$,

$$\sum_{b\neq b'} T_b T_{b'} = \left(\sum_b T_b\right)^2 - B. \tag{A3}$$

For a specified phase-gradient component, the corresponding mixed pair sum satisfies

$$\sum_{s\neq t} Y_s Y'_t = \left(\sum_s Y_s\right)\left(\sum_t Y'_t\right) - \sum_s Y_s Y'_s, \tag{A4}$$

where Y'_s is the phase derivative defined in Sec. III C. These identities allow us to evaluate the objective in Eq. (33) and a specified gradient component in Eq. (36) using $O(S + B)$ arithmetic operations once the sample values are available.

For the sampled-mask estimator in Eq. (42), let $\mathcal{U}$ denote the set of distinct sampled masks and m_u the multiplicity of mask u. Since repeated masks use the same batches, we can write

$$\widehat{\mathrm{MMD}}^2_\sigma(c) = Q^+_\sigma \sum_{u\in\mathcal{U}} \frac{m_u}{M} \widehat{\Delta^2_c}(u). \tag{A5}$$

This regrouping preserves the sampled objective and its gradient while reducing the number of Pauli moments we evaluate.

3. Binomial benchmark

We use the binomial target in Eq. (45) and the product-state phase encoder introduced for this benchmark. The IQP unitary has forty independent parameters on an open 4×4 lattice.

Each of five independent fits uses a fixed pool of 512 target bitstrings at each training condition $c \in \{0, 0.25, 0.5, 0.75, 1\}$, giving 2560 bitstrings per fit. We initialize all forty phases independently from $\mathcal{N}(0, 0.25^2)$.

We perform 1024 `Adam` updates with an initial learning rate of 0.005 and cosine decay to one tenth of this value. At each update, we draw one training condition, 128 target bitstrings with replacement, 1024 reference samples in chunks of 256, and 1024 nonidentity masks. The kernel bandwidth gives expected mask weight one before excluding the identity mask. The remaining optimizer and numerical-precision settings follow Appendix B.

We evaluate the initial parameters, every 32 updates and the final update. Within each fit, we select the checkpoint with the lowest mean exact full-distribution TVD over development conditions $c \in \{0.125, 0.375, 0.625, 0.875\}$. Checkpoint selection therefore uses exact target distributions in addition to the finite training pools.

Reporting uses 80 conditions, $c_j = (j+1/3)/80$ for $j = 0, \ldots, 79$, excluded from training and development selection. We compute exact full-distribution and Hamming-weight profile TVDs, average each metric over reporting conditions within each fit, and report the median across fits. For comparison with the selected-checkpoint results, applying the same reporting procedure to the final checkpoints gives a median full-distribution TVD of 0.0674.

Figure 2(a) shows model profiles averaged across the five fits at four reporting conditions, $c \approx \{0.004, 0.329, 0.667, 0.992\}$, chosen to spread the profiles across the interval. Panel (b) shows the median and interquartile range across fits of the mean Hamming-weight response at all reporting conditions.

4. Curie–Weiss models

We compare four model families for the sixteen-qubit Curie–Weiss target $p(x \mid \beta) \propto \exp[\beta(2|x| - 16)^2/32]$. All models use the same IQP ansatz, with sixteen independent one-body and 120 independent two-body phase parameters,

without ancillary qubits. The final-fitting stage described in Section A 4 a provides the models used in the reported comparisons.

Conditional IQP and temperature-specific IQP use the parity encoder in Eq. (53). The shared product-input model uses $R_Z(s\beta)R_Y(\pi/2)\,|0\rangle$ on each qubit, where s is the encoder scale. The temperature-specific IQP models with standard input use $|0\rangle^{\otimes 16}$ and have no encoder-scale parameter. The product-input comparison retains the same shared IQP ansatz while changing the encoder.

The parity input is invariant under global spin flip, and the IQP unitary preserves this invariance for every parameter vector. This guarantees a symmetric magnetization profile but does not impose permutation symmetry or agreement with the target distribution.

To prepare the parity input, we initialize all sixteen qubits in $|0\rangle$, apply $R_X(s\beta)$ to each of the first fifteen qubits, and apply a CNOT from each of these qubits to qubit sixteen. Before the final Hadamard layer, qubit sixteen records the parity of the first fifteen bits. This state is the reference state $|\psi_{\mathrm{P}}\rangle = H^{\otimes 16}\,|\phi_{\mathrm{P}}\rangle$ used for classical estimation. Applying the Hadamard layer produces the encoded input.

For a reference bitstring (r, b), with $r \in \{0,1\}^{15}$ and $b \in \{0,1\}$, the computational-basis amplitude is

$$\psi_{\mathrm{P}}(r,b) = \mathbf{1}_{b=\oplus_i r_i}\cos(s\beta/2)^{15-|r|}[-i\sin(s\beta/2)]^{|r|}. \tag{A6}$$

The indicator restricts the reference state to configurations whose final bit equals the parity of the first fifteen bits.

We sample the reference distribution by drawing each of the first fifteen bits independently with probability $\sin^2(s\beta/2)$ of being one, then setting the final bit to their parity. The product expression gives the required amplitude ratios, with denominators evaluated only at sampled configurations with nonzero amplitude. Any odd-parity mask maps a supported configuration outside the reference support and therefore gives a zero ratio. Sampling and amplitude-ratio evaluation each require $O(n)$ arithmetic operations for the corresponding n-qubit construction.

We organize fitting into three stages. Screening compares hyperparameter configurations, confirmation refits a shortlist using fresh repetitions, and final fitting produces the models used for reporting. A configuration specifies the optimizer, initialization, kernel and applicable encoder settings. Each repetition fits one shared circuit or a family of three temperature-specific circuits, using the prescribed number of initializations per circuit.

We fit the conditional IQP and shared product-input models at $\beta \in \{0.1, 0.7, 1.3, 1.95\}$. Each temperature-specific family contains three circuits fitted directly at $\beta \in \{0.4, 1, 2\}$. These circuits have separate learned phases but use one jointly selected hyperparameter configuration.

Each training temperature supplies a fixed pool of 1024 target bitstrings per repetition. We reuse these pools across updates and initializations. Within each stage, we pair compatible target pools and stochastic streams across configurations and matched model families; the three stages use independent streams. Each update draws 256 target bitstrings with replacement, 2048 nonidentity masks and 1024 reference samples. Shared models sample each training temperature with probability 1/4. We sample the three bandwidth entries with equal probability, retaining repeated entries. Table IV summarizes the training budgets.

TABLE IV. Curie–Weiss training budgets. Initializations are counted per circuit and repetition; the final column totals optimizer updates across all repetitions and initializations. All stages use 2000 updates per initialization, with checkpoint evaluation at initialization and every 100 updates. Each shared-model repetition fits one circuit using 4096 target-pool entries. Each temperature-specific-family repetition fits three circuits using 3072 entries, and therefore uses three times the learned phases and optimizer updates.

Stage	Repetitions	Initializations	Total updates per circuit
Screening, per proposal	5	1	10,000
Confirmation, per candidate	5	3	30,000
Final fitting, per family	10	3	60,000

a. Hyperparameter optimization

We select checkpoints and hyperparameter configurations using a full-distribution TVD score J and a physical-observable score P. For a temperature grid G and a nonnegative error curve e, define

$$R_G(e) = \frac{2\max_{\beta\in G} e(\beta) + \overline{e}_G}{3}, \tag{A7}$$

where $\bar{e}_G$ is the trapezoidal integral over the grid divided by its temperature span. This score gives the maximum error twice the weight of the temperature-averaged error. For a single temperature, both terms equal the error at that temperature. We compute $J = R_G(\mathrm{TVD})$ from exact probabilities over all 2^{16} bitstrings.

The physical-observable score combines errors in mean absolute magnetization and energy fluctuations,

$$P = \frac{R_G(e_m) + R_G(e_c)}{2}, \qquad e_m = \left|\langle|m|\rangle_q - \langle|m|\rangle_p\right|, \tag{A8}$$

with $e_c = |c_{\mathrm{fluc},q} - c_{\mathrm{fluc},p}|/c_\star$ and the fixed target-only normalization $c_\star \approx 0.823110$. Section A 4 d describes these observables and their interpretation. Signed mean magnetization and departures from spin-flip invariance remain diagnostics and do not enter selection.

For shared models, checkpoint selection uses the development grid

$$\begin{aligned} A = &\{0.05 + 0.12j : j = 0, \ldots, 15\} \\ &\cup \{1.975\}. \end{aligned} \tag{A9}$$

Within each repetition, we pool all scheduled checkpoints across the circuit's initializations and find the minimum score $J_{\min}$. We then select the checkpoint with the smallest P among those satisfying $J \le J_{\min} + 0.01$. Ties favor smaller J, then earlier updates and lower initialization index. The tolerance applies to the complete checkpoint pool and does not imply statistical equivalence. Temperature-specific models use the same rule at their directly fitted temperatures.

We assign one `Optuna` study to each model family [41]. Each study evaluates 60 independent random configurations followed by 60 adaptive proposals from a tree-structured Parzen estimator (TPE), giving 120 screening configurations per family. TPE uses the random-stage observations from the same study. Table V specifies the search space, including the categorical representation of the bandwidth pair selected during earlier development.

Under the specified random-search distribution, 60 independent proposals have probability $1 - 0.95^{60} \approx 0.954$ of including at least one configuration in its best-performing 5% according to the screening objective [42]. This is a coverage statement about the search distribution, not a bound on error or proximity to a global optimum. It does not apply to the subsequent 60 adaptive TPE proposals.

TABLE V. Curie–Weiss hyperparameter search space and initial random sampling distributions. TPE adapts its proposals after the random stage. We use `Adam` for all models. The encoder scale applies to the parity and product encoders; the standard-input family has no encoder-scale parameter.

Parameter	Options or range	Initial prior
Learning rate	$[3 \times 10^{-4}, 10^{-1}]$	Log-uniform
Width s_θ	$[0.01, 1]$ radians	Log-uniform
b_1	$[0.8, 0.95]$	Uniform
$1 - b_2$	$[0.001, 0.05]$	Log-uniform
Schedule	Constant, cosine	Equal probabilities
Encoder scale s	$[0.25, 1.5]$	Log-uniform
Kernel pair (w_2, w_3)	45 ordered pairs	Uniform categorical

The expected mask-weight entries are $(2, w_2, w_3)$, where $w_2 \le w_3$ and $w_2, w_3 \in \{4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 7.99\}$. Repeated entries are allowed and retain their sampling weight.

We fix $\epsilon = 10^{-8}$ and weight decay to zero, without warmup or gradient clipping. The cosine schedule decreases the learning rate to one tenth of its initial value over 2000 updates. We draw initial phases from the specified Gaussian without clipping or wrapping. The encoder-scale range gives $s\beta \le 3 < \pi$ over $\beta \in [0, 2]$; this bound applies to the encoder angles, not the Gaussian circuit phases. We use multivariate TPE with 60 startup observations and 24 expected-improvement candidates, without pruning or queued reference configurations.

A numerical failure consumes a proposal and receives a search penalty of 1, which is not a measured TVD. Invalid configurations cannot enter confirmation. All 480 screening proposals were valid, and confirmation and final fitting had no numerical failures.

During screening, shared models use A for checkpoint selection and configuration scoring. For each temperature-specific family, we combine the three selected circuits' errors using R_G within each repetition. We then average J and P across the five repetitions for each configuration. Shared-model scores likewise average over repetitions. TPE receives only J; we retain P and the component error curves separately.

Confirmation refits three or four distinct configurations from each study. The shortlist includes the configuration with minimum J, the minimum-P configuration within $J_{\min} + 0.01$, and further configurations ranked by J until it

TABLE VI. Hyperparameter configurations selected by confirmation. Each temperature-specific family's settings apply jointly to three circuits with separate learned phases. Other optimizer and training settings follow the protocol described in this subsection. Displayed continuous values are rounded; machine-readable settings retain full precision.

Parameter	Conditional IQP	Shared product-input IQP	Temperature-specific IQP	Temperature-specific IQP, standard input
Initial learning rate	0.00341745	0.0300696	0.00430287	0.0898283
Width s_θ (radians)	0.0221053	0.0454769	0.0300356	0.0326103
`Adam` b_1	0.801879	0.826492	0.899094	0.946166
`Adam` b_2	0.991964	0.998983	0.96613	0.998814
Schedule	cosine	constant	constant	cosine
Encoder scale s	0.683987	0.657269	0.582332	–
Expected kernel weights	[2, 4, 7.5]	[2, 4, 5]	[2, 7.5, 7.99]	[2, 5, 5.5]

contains at least three. We also include the global minimum-P configuration if it is not already present. This gives three conditional IQP configurations and four configurations in each other family. Each receives five fresh repetitions with three initializations per circuit.

Shared-model confirmation continues to select checkpoints on A, but ranks configurations on $A \cup B$, where

$$\begin{aligned} B = \{&0.075, 0.135, 0.23, 0.35, 0.47, 0.59, \\ &0.675, 0.735, 0.83, 0.95, 1.07, 1.19, \\ &1.275, 1.335, 1.43, 1.55, 1.67, 1.79, 1.9, 1.9625\}. \end{aligned} \tag{A10}$$

Temperature-specific confirmation uses the directly fitted temperatures. We select the configuration with the smallest confirmed P among those within 0.01 of the minimum confirmed J. Confirmation results do not generate further TPE proposals.

After fixing one configuration per family, we perform ten fresh final repetitions with three initializations per circuit. Checkpoint selection continues to use A for shared models and the fitted temperature for each temperature-specific circuit. We retain all ten selected outcomes, compute each reporting metric within each repetition, and then report means and sample standard deviations using denominator 9.

The shared-model reporting grid contains 101 points: 0, 2 and $(6j-4)/297$ for $j = 1, \ldots, 99$. We also evaluate the shared models at the common temperatures $\beta \in \{0.4, 1, 2\}$. The reporting grid and these temperatures are excluded from shared-model training and selection. Temperature-specific models provide comparisons only at their directly fitted temperatures. Earlier model development explored this temperature range, and evaluation on a finite grid does not establish accuracy throughout the continuous interval.

We use double precision for circuit parameters and exact probabilities, while matrix products in the sampled estimator use single precision. Exact probability normalization uses absolute tolerance 10^{-8} and zero relative tolerance. Table VI reports the selected configurations.

b. *Product-input ablation*

We test how encoder choice affects performance while retaining the shared IQP architecture and training protocol. We compare the conditional IQP model, which uses the parity encoder, with a shared product-input IQP model whose encoder prepares

$$|\phi_{\mathrm{prod}}(\beta; s)\rangle = \bigotimes_{i=1}^{16} R_Z(s\beta) R_Y(\pi/2) \, |0\rangle_i \, . \tag{A11}$$

Both encoders map temperature to input phases and produce uniform computational-basis input probabilities. The product encoder does not enforce global spin-flip symmetry. However, replacing the parity encoder also changes the input correlations and phase generators, so differences in distributional accuracy cannot be attributed solely to symmetry.

Both models use 136 independent IQP phases, the same training temperatures and hyperparameter search ranges, and the same search and training budgets per circuit. We pair compatible stochastic streams and target pools across the two models. Each encoder has a separate hyperparameter study and selected configuration, as reported in Table VI. We evaluate both models on the same reporting conditions using ten fresh final repetitions.

TABLE VII. Comparison of conditional IQP with the parity encoder and shared product-input IQP. The three temperature columns report exact full-distribution TVD. The aggregate scores $J = R_G(\mathrm{TVD})$ and $P = [R_G(e_m) + R_G(e_c)]/2$ use the same 101-point reporting grid G, with R_G defined in the hyperparameter-optimization protocol. Values are means ± sample standard deviations over ten fresh final repetitions. Lower values indicate smaller errors. The comparison changes input correlations and encoder phase generators as well as enforced spin-flip symmetry.

Model	$\beta = 0.4$	$\beta = 1$	$\beta = 2$	J	P
Conditional IQP	0.1185 ± 0.0107	0.1396 ± 0.0048	0.0832 ± 0.0103	0.1281 ± 0.0027	0.0865 ± 0.0088
Shared product-input IQP	0.1576 ± 0.0079	0.2242 ± 0.0113	0.1104 ± 0.0169	0.2051 ± 0.0088	0.1393 ± 0.0107

The parity encoder gives lower aggregate errors in this comparison. Relative to shared product-input IQP, conditional IQP reduces the mean full-distribution TVD score J by 37.5% and the mean physical-observable score P by 37.9%, as reported in Table VII. Its mean full-distribution TVD is also lower at every nonzero point of the reporting grid. The temperature-averaged TVD is 0.1033 ± 0.0036 for conditional IQP and 0.1662 ± 0.0063 for shared product-input IQP; the corresponding per-repetition maximum TVDs are 0.1406 ± 0.0037 and 0.2245 ± 0.0112. Figure 8 shows that the energy-fluctuation error curves nevertheless cross. A lower aggregate P therefore does not imply lower error for every observable at every temperature.

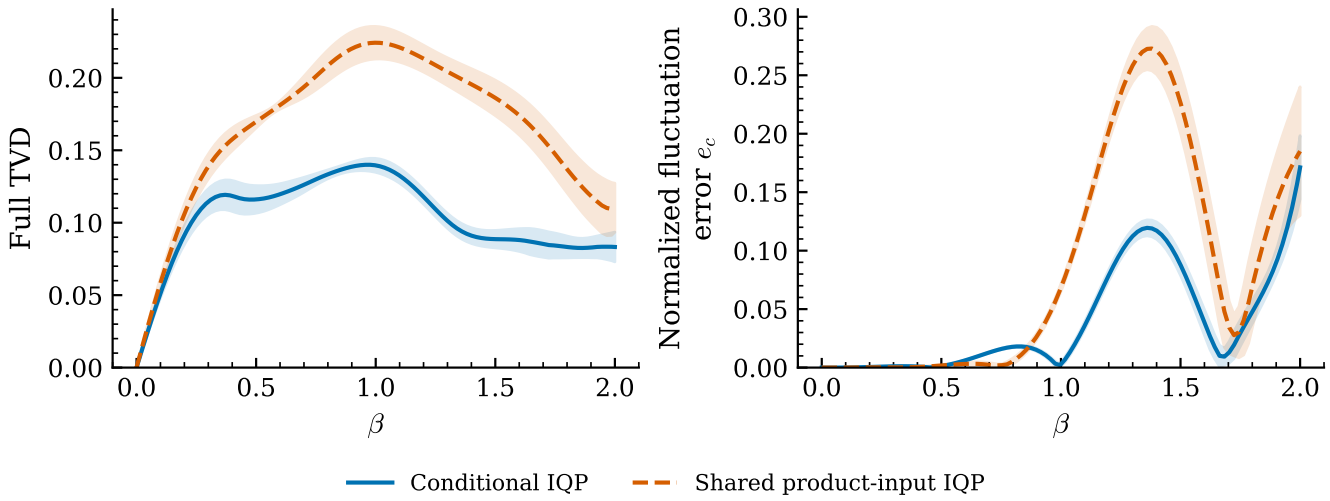


FIG. 8. Encoder comparison for shared conditional IQP models on the 101-point reporting grid. Solid blue curves show conditional IQP with the parity encoder; dashed orange curves show shared product-input IQP. Curves and shaded bands represent means and sample standard deviations over ten fresh final repetitions. The left panel shows exact full-distribution TVD; the right shows the normalized energy-fluctuation error $e_c = |c_{\mathrm{fluc},q} - c_{\mathrm{fluc},p}|/c_\star$, with $c_\star \approx 0.823110$. Lower values indicate smaller errors. Both models use the same training temperatures and search and training budgets per circuit, with separately selected hyperparameter configurations. Both architectures produce the exact uniform target distribution at $\beta = 0$ for every parameter vector.

c. *Profile and full-distribution errors*

Table VIII reports profile TVD from the same ten fresh final repeats as the full-distribution comparison. We compute the profile error in each repeat before averaging, rather than the error of the averaged profile. At $\beta = 0.4$, the temperature-specific IQP model with standard input has mean profile TVD 0.0281 ± 0.0247 but full TVD 0.3787 ± 0.0034, exposing errors within Hamming-weight shells. At $\beta = 1$, temperature-specific IQP has smaller profile TVD than conditional IQP, 0.0642 ± 0.0028 versus 0.1203 ± 0.0046, although its full TVD is larger. Hamming-weight profiles and full distributions can therefore give different accuracy rankings, which is why full-distribution TVD remains our primary accuracy metric.

TABLE VIII. Exact Hamming-weight profile TVD at sixteen qubits. Values are means ± sample standard deviations over ten fresh final repeats, computed within each repeat before aggregation. Lower is better. Conditional IQP uses the shared parity encoder and excludes markers from fitting and selection; the temperature-specific IQP models use the parity encoder or the standard input $|0\rangle^{\otimes 16}$ and fit directly at each marker.

Model	$\beta = 0.4$	$\beta = 1$	$\beta = 2$
Conditional IQP	0.0438 ± 0.0040	0.1203 ± 0.0046	0.0646 ± 0.0124
Temperature-specific IQP	0.0149 ± 0.0065	0.0642 ± 0.0028	0.1298 ± 0.0065
Temperature-specific IQP, standard input	0.0281 ± 0.0247	0.0833 ± 0.0158	0.4003 ± 0.1736

TABLE IX. Physical-observable errors for sixteen-qubit Curie–Weiss models at three common evaluation temperatures. The mean absolute magnetization error is $e_m = |\langle|m|\rangle_q - \langle|m|\rangle_p|$, and the normalized energy-fluctuation error is $e_c = |c_{\mathrm{fluc},q} - c_{\mathrm{fluc},p}|/c_\star$, with $c_\star \approx 0.823110$. Values are means ± sample standard deviations over ten fresh final repetitions, with errors computed separately within each repetition before averaging. Lower values indicate smaller errors. Both shared models exclude these temperatures from training and selection. Temperature-specific models fit directly at each listed temperature, using either the parity encoder or the standard input $|0\rangle^{\otimes 16}$.

Model	Error	$\beta = 0.4$	$\beta = 1$	$\beta = 2$
Conditional IQP	e_m	0.0167 ± 0.0017	0.0581 ± 0.0028	0.0083 ± 0.0043
Conditional IQP	e_c	0.0009 ± 0.0002	0.0028 ± 0.0015	0.1712 ± 0.0272
Shared product-input IQP	e_m	0.0174 ± 0.0039	0.0834 ± 0.0103	0.0130 ± 0.0066
Shared product-input IQP	e_c	0.0005 ± 0.0004	0.0683 ± 0.0034	0.1851 ± 0.0551
Temperature-specific IQP	e_m	0.0006 ± 0.0004	0.0023 ± 0.0018	0.0127 ± 0.0013
Temperature-specific IQP	e_c	0.0013 ± 0.0003	0.0475 ± 0.0042	0.0705 ± 0.0053
Temperature-specific IQP, standard input	e_m	0.0022 ± 0.0028	0.0030 ± 0.0024	0.2759 ± 0.1435
Temperature-specific IQP, standard input	e_c	0.0003 ± 0.0002	0.0036 ± 0.0023	2.8935 ± 1.4896

d. Physical observables

For a distribution q, let $\rho_q(k \mid \beta)$ denote its exact Hamming-weight profile and set $m_k = (2k - n)/n$ with $n = 16$. We compute its moments as $\langle m^r\rangle_q = \sum_{k=0}^{n} \rho_q(k \mid \beta) m_k^r$, and its mean absolute magnetization as $\sum_k \rho_q(k \mid \beta)|m_k|$. The target law corresponds to $H = -nm^2/2$ in units $J = k_B = 1$. Excluding self-interactions would add only a constant to H and would not change the energy variance. Equation (55) therefore becomes

$$c_{\mathrm{fluc},q}(\beta) = \frac{\beta^2 n}{4}\left(\langle m^4\rangle_q - \langle m^2\rangle_q^2\right). \tag{A12}$$

The observables depend only on the shell profile and do not test uniformity within shells. For the Gibbs target, differentiating the mean energy per spin with respect to $T = \beta^{-1}$ gives this fluctuation expression. We do not assume that the learned family satisfies the same derivative identity.

We evaluate ten selected circuits per shared model on the fixed reporting grid, keeping their parameters fixed after checkpoint selection on A. Temperature-specific models provide comparisons at their three fitted temperatures. We compute each observable and its absolute target error separately within each repetition, then report means and sample standard deviations. In particular, we compute the energy-fluctuation statistic before averaging across repetitions.

Table IX reports e_m and e_c at the common evaluation temperatures for all four model families. On these temperatures, the aggregate physical-observable score P is 0.0914 ± 0.0102 for conditional IQP, 0.1130 ± 0.0223 for shared product-input IQP, 0.0363 ± 0.0018 for temperature-specific IQP and 1.2223 ± 0.6289 for temperature-specific IQP with standard input. These scores use only the three common temperatures and therefore differ from the shared-model scores over the full reporting grid in Table VII. The standard-input model shows substantial variation across repetitions at $\beta = 2$. We retain all ten outcomes in the reported statistics.

At $\beta = 0$, both shared encoders produce the exact uniform target distribution for every IQP parameter vector. Its Hamming-weight profile is $\rho_0(k) = 2^{-n}\binom{n}{k}$, so agreement at this endpoint follows from the architecture independently of training.

5. Scaling

We use the parity encoder in Eq. (53), generalized to n visible qubits, without ancillas. Its selected scale is $s = 1.18430$, and it preserves global spin-flip symmetry at every size. The learned unitary uses the open rectangular lattice architectures in Table X. At size n, the target distribution is $p(x \mid \beta) \propto \exp[\beta(2|x| - n)^2/(2n)]$.

We train at $\beta \in \{0.6, 0.8, 1.2, 1.4\}$, using fixed pools of 10,000 target samples per condition. We draw a separate validation pool of 2000 targets per training condition. Held-out temperatures $\beta \in \{0.4, 0.95, 1.0, 1.05, 1.6\}$ enter reporting only and do not select checkpoints or hyperparameters.

TABLE X. Open rectangular lattice architectures for the scaling experiments. We assign one trainable phase to each qubit and one to each horizontal or vertical nearest-neighbor pair.

Visible qubits	Lattice	Trainable parameters
64	8×8	176
128	8×16	360
256	16×16	736
512	16×32	1,488
1,024	32×32	3,008
2,048	32×64	6,048

We select checkpoints using four fixed collections of mask and reference draws, which we call validation banks. Each bank contains 512 nonidentity mask draws per condition and bandwidth, and 1024 reference draws per condition, shared across bandwidths. All four banks use the same validation target pools. We reuse the banks across checkpoints and pair compatible stochastic streams across candidates within each search stage. Let $\overline{M}_n^2(\boldsymbol{\theta}; \beta)$ denote the validation MMD^2 estimate averaged over the four banks and two bandwidths. We select checkpoints using

$$V_n(\boldsymbol{\theta}) = \frac{1}{4} \sum_{\beta \in \{0.6, 0.8, 1.2, 1.4\}} \frac{\overline{M}_n^2(\boldsymbol{\theta}; \beta)}{d_{n,\beta}}. \tag{A13}$$

Each fixed scale $d_{n,\beta}$ is the median independently evaluated training-temperature MMD^2 across three earlier baseline runs, floored at 10^{-3}. These candidate-independent scales balance the temperatures' contributions to selection. Training continues to sample conditions and bandwidths uniformly, and reporting uses unnormalized MMD^2. We retain negative unbiased estimates.

We optimize the six hyperparameters in Table XI. The first 16 proposals comprise a baseline configuration, a proposal from earlier sixteen-qubit parity-model optimization, and 14 random proposals. The remaining 32 proposals use a tree-structured Parzen estimator.

TABLE XI. Scaling hyperparameter search space and initial random sampling distributions.

Hyperparameter	Range	Initial random sampling
Initial learning rate	$[10^{-4}, 0.05]$	Log-uniform
Initialization width s_θ (radians)	$[0.005, 0.3]$	Log-uniform
Encoder scale s	$[0.25, 1.5]$	Uniform
b_1	$[0.8, 0.95]$	Uniform
b_2	$[0.95, 0.999]$	Uniform
Final-to-initial learning-rate ratio	$[0.05, 1]$	Log-uniform

Screening evaluates each of the 48 proposals at 64 and 256 qubits, with one initialization and 2000 updates per size. Each proposal's score averages its minimum checkpoint-selection score across the two sizes. We confirm the three leading proposals using three fresh runs per size, one initialization per run, and 5000 updates. We also evaluate the baseline under the same confirmation budgets. The lowest mean confirmation score determines the configuration used at all six sizes.

Final fits use the settings in Table III, with three runs per size and three initializations of 5000 updates per run. Each starting parameter vector has phases drawn independently from $\mathcal{N}(0, s_\theta^2)$. The runs use separate target pools and stochastic training streams. However, their initialization sets overlap, giving five distinct starting parameter vectors per system size. Within each run, the initializations reuse target pools and stochastic training streams, and we reset the optimizer for each initialization. We evaluate the initial parameters and every 100 updates, then select the lowest-scoring checkpoint across all three initializations.

For each run, we report the selected initialization's trajectory at update 0 and every 500 updates through update 5000. Reporting uses five independently drawn evaluation banks. Each bank contains 2000 target draws per held-out condition, 512 mask draws per condition and bandwidth, and 1024 reference draws per condition. We share reference samples across bandwidths and reuse each bank across reporting updates. These draws are independent of training and checkpoint selection. At each update, we average the estimated MMD^2 over the five banks, held-out temperatures and two bandwidths to obtain one value per run. Figure 6 shows the median and interquartile range of these values across the three runs. These summaries combine variation from training and sampled evaluation.

Appendix B: Two-coordinate encoder and evaluation protocol

1. Training, development and reporting conditions

We use the binomial-centered target family in Eq. (50) and the phase encoder in Eq. (51). Training conditions determine the sampled MMD objective, development conditions select hyperparameters and checkpoints, and reporting conditions assess the selected models. Axis training, presented in the main text, tests new combinations of the two coordinates, both inside and outside the training convex hull $|u| + |v| \le 1$. As a complementary experiment, grid training tests generation at new conditions within the square training convex hull $[-1, 1]^2$.

Axis training uses $(u, 0)$ at $u \in \{-1, -0.37, 0, 0.37, 1\}$ and $(0, v)$ at $v \in \{-1, -0.31, 0.31, 1\}$. Development uses $(u, 0)$ at $u \in \{-0.73, -0.17, 0.17, 0.73\}$ and $(0, v)$ at $v \in \{-0.69, -0.13, 0.13, 0.69\}$. Reporting uses sixteen mixed pairs,

$$\{-1, -0.37, 0.37, 1\} \times \{-1, -0.31, 0.31, 1\}.$$

Four lie inside the training convex hull $|u| + |v| \le 1$, and twelve lie outside. Each individual coordinate value appears in training, but no mixed reporting pair appears in training or development. Earlier exploratory work examined these condition sets, so the benchmark assesses the fitted models within an already studied domain.

Grid training uses $\{-1, 0, 1\}^2$, with development conditions

$$\{-1, -0.5, 0, 0.5, 1\} \times \{-0.75, -0.25, 0.25, 0.75\}.$$

The reporting grid is

$$u \in \{-1, -0.83, -0.37, 0.19, 0.61, 1\},$$
$$v \in \{-1, -0.79, -0.31, 0.23, 0.67, 1\}.$$

We exclude its four training corners, leaving thirty-two reporting pairs.

2. Sampled training and exact development selection

For each training design, we define the development score as the mean exact full-distribution TVD over its development conditions. We use this score to select checkpoints and hyperparameter configurations. We evaluate the initial parameters, every 64 updates during screening or 100 updates during confirmation and final fitting, and the final update. Each fit retains the checkpoint with the lowest development score.

Each fit uses a fixed pool of 1024 target bitstrings at each of nine training conditions. An update draws one condition and one bandwidth uniformly, 256 target strings with replacement, 1024 fresh reference samples and 1024 nonidentity masks. Expected mask weights before conditioning on nonidentity are $\{2, 4\}$, with bandwidth $\sigma = (-2\ln(1-2w/16))^{-1/2}$ and the identity-mask correction. Conditional on the target pools, the sampled MMD estimates the objective for their empirical distributions.

We use `Adam` with $b_1 = 0.9$, $b_2 = 0.999$, $\epsilon = 10^{-8}$ and zero weight decay. Cosine decay reduces the learning rate to one tenth of its initial value. Estimator accumulation uses double precision, matrix products use single precision, and reference batches use chunks of 256.

For each training design, we screen four hyperparameter configurations for 512 updates. These combine initial learning rates 0.01 and 0.03 with one-body initialization means $m = 0$ and $m = \pi/3$. The sixteen one-body parameters are independent draws from $\mathcal{N}(m, 0.1^2)$, while the twenty-four pair parameters are independent draws from $\mathcal{N}(0, 0.1^2)$. We retain the two configurations with the lowest development scores and perform two confirmation fits per configuration, each with 2000 updates. The lower mean confirmation development score determines the selected configuration.

Both training designs select $m = \pi/3$ and learning rate 0.01. We fix these configurations before reporting evaluation, then perform five independent final fits per design, each with 2000 updates. Screening, confirmation and final fitting total twenty-six fits. We pair compatible target pools and stochastic streams across configurations and training designs within each repetition. Final repetitions are independent of configuration selection.

Model selection uses exact target distributions at development conditions in addition to finite training pools. The target-family design and encoder phase coefficients also reflect earlier exploratory work. All ten final fits select a checkpoint after at least 1000 updates. Initialization with one-body mean $m = \pi/3$ gives lower development scores in this search, so the campaign does not establish robustness across initialization regimes.

Independent checks compare reference samples, amplitude ratios, circuit probabilities and sampled losses and gradients with dense or public circuit calculations. At four qubits, 512 independent sampled estimates agree with an exact empirical-pool MMD calculation within 0.67 standard errors for the loss and 1.43 standard errors for every gradient coordinate. A sixteen-qubit check at three condition pairs agrees with direct circuit probabilities to below 6×10^{-18}. These checks test implementation agreement at the stated settings, rather than bounding estimator tails throughout the domain.

3. Reference access and circuit preparation

The encoded input is invariant under qubit permutations, and the Hadamard layer preserves this symmetry. Input and reference amplitudes therefore depend only on Hamming weight. For an input bitstring of weight k, the amplitude is

$$a_{u,v}(k) = 2^{-8} \exp\left[\frac{iu(k-8)}{2} - \frac{iv(k-8)^2}{40}\right]. \tag{B1}$$

We write $\psi_{u,v}(j)$ for the amplitude of any reference bitstring of weight j. Grouping the terms in the Hadamard transform by input weight k and overlap ℓ gives

$$\psi_{u,v}(j) = 2^{-8} \sum_{k=0}^{16} a_{u,v}(k) \mathcal{K}_k(j), \tag{B2}$$

$$\mathcal{K}_k(j) = \sum_{\ell=0}^{k} (-1)^\ell \binom{j}{\ell} \binom{16-j}{k-\ell}, \tag{B3}$$

where ℓ counts positions occupied in both bitstrings, and binomial coefficients outside their valid ranges are zero.

We tabulate the seventeen reference amplitudes. To sample a reference bitstring, we first draw its weight j with probability $\binom{16}{j}|\psi_{u,v}(j)|^2$, then draw a bitstring uniformly among those with weight j. For a reference sample z and bit mask b, the amplitude ratio is

$$R_{u,v}(z,b) = \frac{\psi_{u,v}(|z \oplus b|)}{\psi_{u,v}(|z|)}. \tag{B4}$$

We evaluate this ratio only at sampled configurations with nonzero reference amplitude.

At $(u,v) = (0,0)$, the reference state is exactly $|0\rangle^{\otimes 16}$. Extending the weight-based construction to general n requires a polynomial number of arithmetic operations for reference sampling and amplitude-ratio evaluation. Efficient access nevertheless does not ensure that the chosen sample budgets resolve small losses or gradient components.

We prepare the encoded input using sixteen Hadamard gates, sixteen $R_Z(u/2)$ rotations and 120 pairwise ZZ rotations. Since

$$J_z^2 = \frac{n}{4} I + \frac{1}{2} \sum_{i<j} Z_i Z_j,$$

applying $\mathrm{IsingZZ}_{ij}(v/40) = e^{-ivZ_iZ_j/80}$ to every pair implements the quadratic phase up to a global phase. This preparation requires no ancillas or target-amplitude preparation. The diagonal IQP layer contains sixteen one-body and twenty-four nearest-neighbor two-body phases with independent parameters. The complete IQP unitary also includes the two Hadamard layers in Eq. (2). These are high-level gate counts; the experiment does not assess compiled depth or deployment noise.

TABLE XII. Two-coordinate reporting errors. Values are means and sample standard deviations of per-fit mean errors across five independent fits. Full TVD is the primary metric and selects checkpoints on separate development conditions. The two fitting geometries have different reporting sets, so columns do not form a paired accuracy comparison. Lower is better.

	Axis training, sixteen pairs	Grid training, thirty-two pairs
Encoder only, full and profile TVD	0.3867	0.3586
Learned model, full TVD	0.0468 ± 0.0143	0.0397 ± 0.0093
Learned model, profile TVD	0.0173 ± 0.0066	0.0156 ± 0.0036

4. Full-distribution evaluation and response diagnostics

The reference state is permutation symmetric, but the independently parameterized lattice generally breaks this symmetry. We therefore compute all 2^{16} output probabilities for evaluation and use full-distribution TVD as the primary accuracy metric. Hamming-weight profile TVD may hide errors within shells. These exact calculations provide small-system diagnostics alongside the sampled MMD used for training.

Table XII reports the axis-training results and the complementary grid-training results. Grid training gives a selected full-distribution TVD of 0.0397 ± 0.0093 over thirty-two held-out pairs. The two designs have similar error patterns on the common 41×41 diagnostic grid. Their condition-averaged mean TVDs are 0.0337 for axis training and 0.0318 for grid training. These sweep averages include training and development conditions and are not held-out reporting scores. The small difference gives limited evidence of an additional benefit from joint training conditions in this family; we do not interpret it as evidence that extrapolation has no cost.

We average reporting errors over conditions within each fit, then report means and sample standard deviations across five fits. The two training designs use different reporting sets, so their average errors do not provide a direct comparison of accuracy. At the final update, the mean full-distribution TVDs are 0.0489 ± 0.0137 for axis training and 0.0412 ± 0.0087 for grid training. We report these values separately from the development-selected results.

After model selection, we evaluate all 1681 pairs on $u, v \in \{-1, -0.95, \ldots, 0.95, 1\}$. Figure 3 shows the arithmetic mean of the five fits' full-distribution TVDs at each condition, without smoothing. We average the errors rather than first averaging the model distributions. This diagnostic sweep includes training and development conditions.

We also examine how closely the models reproduce changes in mean and width along fixed-coordinate slices. Figures 9 and 10 show these responses. To quantify their magnitude, we divide each model's mean change from $u = -1$ to $u = 1$, or standard-deviation change from $v = -1$ to $v = 1$, by the corresponding exact target change. A response ratio of one indicates agreement in the endpoint change.

Averaged across five fits, the mean-change ratios at $v = (-1, 0, 1)$ are $(0.966, 1.000, 1.043)$ for axis training and $(0.964, 0.998, 1.042)$ for grid training. The corresponding width-change ratios at $u = (-1, 0, 1)$ are $(0.917, 0.829, 0.917)$ and $(0.898, 0.814, 0.898)$. Thus, the average endpoint mean changes closely match the targets, while the average width changes are smaller.

Individual fits recover between 0.507 and 1.073 of the target width change. None reverses an adjacent width step on the tested sweeps. These diagnostics measure response magnitude on selected slices and do not establish accurate calibration throughout the domain. Agreement at $(u, v) = (0, 0)$ holds for every parameter vector and is excluded from the separate reporting averages.

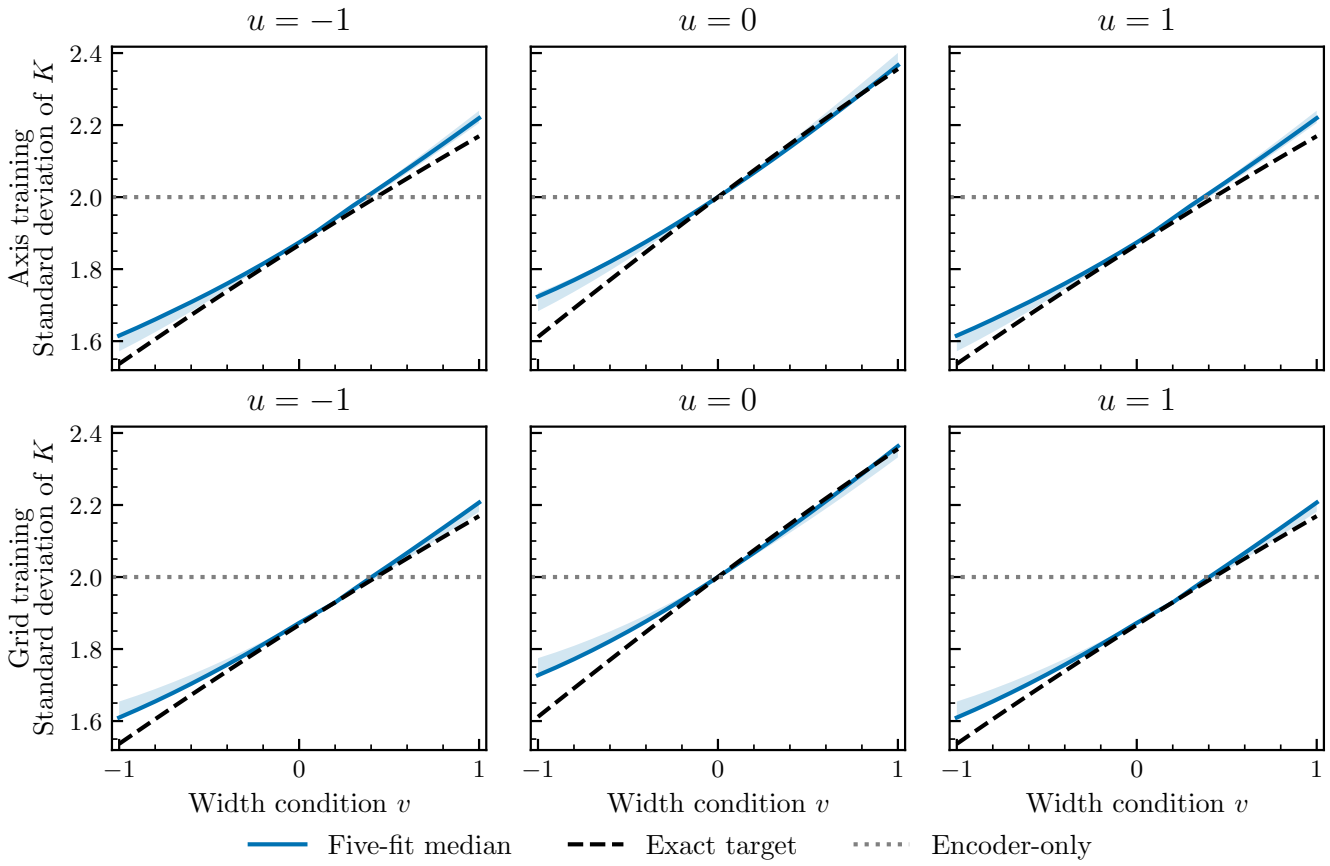


FIG. 9. Width response of sixteen-qubit conditional IQP models, measured by the standard deviation of Hamming weight K. Columns fix $u = -1, 0, 1$, while the horizontal axis varies the width condition v. The top row shows axis training and the bottom row shows grid training. Solid blue curves and shading show the pointwise median and interquartile range across five development-selected fits. Dashed black curves show exact target standard deviations, and dotted gray lines show the encoder-only binomial standard deviation. Each slice contains 21 conditions. These diagnostics follow model selection and include previously examined conditions.

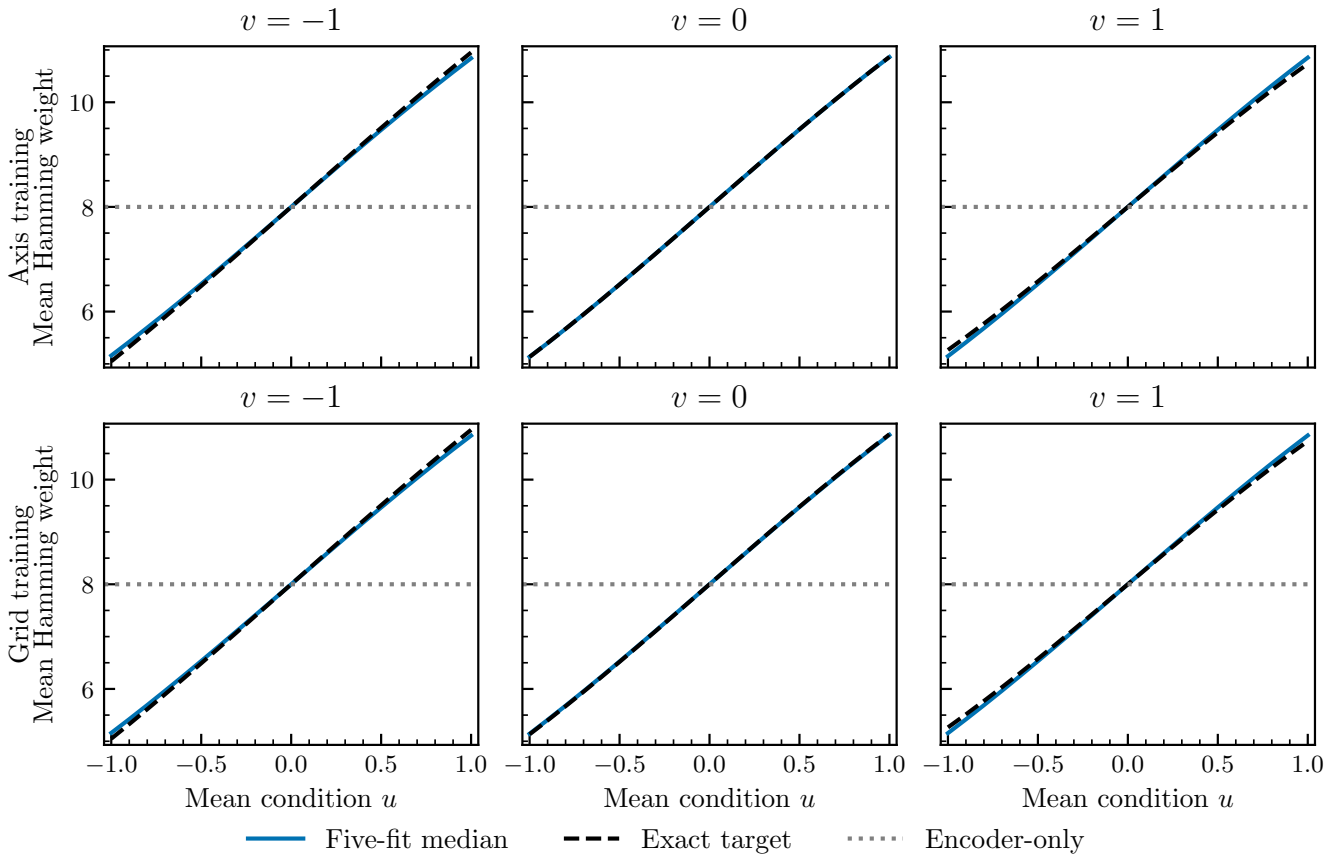


FIG. 10. Mean Hamming-weight response of sixteen-qubit conditional IQP models. Columns fix $v = -1, 0, 1$, while the horizontal axis varies the mean condition u. The top row shows axis training and the bottom row shows grid training. Solid blue curves and shading show the pointwise median and interquartile range across five development-selected fits. Dashed black curves show exact target means, and dotted gray lines show the encoder-only mean of eight. Each slice contains 21 conditions. The target mean depends on both coordinates. These diagnostics follow model selection and include previously examined conditions.

Appendix C: Binary-linear conditional encoders

We use the construction in Section V, with a full-column-rank matrix $C \in \mathbb{F}_2^{n\times k}$ and normalized factor amplitudes $f_\ell(0;c)$ and $f_\ell(1;c)$. All matrix operations below are over $\mathbb{F}_2$. Since C is injective, distinct latent strings u give orthogonal computational-basis states $|Cu\rangle$. The reference-state norm therefore satisfies

$$\begin{aligned}\langle\psi_c|\psi_c\rangle &= \sum_{u\in\mathbb{F}_2^k}\prod_{\ell=1}^{k}|f_\ell(u_\ell;c)|^2 \\ &= \prod_{\ell=1}^{k}\left(|f_\ell(0;c)|^2+|f_\ell(1;c)|^2\right)=1.\end{aligned} \tag{C1}$$

The probability of a reference configuration Cu is $\prod_\ell |f_\ell(u_\ell;c)|^2$. We can therefore sample the reference state by drawing the latent bits independently and returning Cu.

For amplitude-ratio evaluation, we precompute a left inverse $L \in \mathbb{F}_2^{k\times n}$ satisfying $LC = I_k$. Given a reference sample z, we recover its latent coordinates as $u = Lz$. For a mask a, we compute $d = La$ and test whether $Cd = a$. If the test fails, $a \notin \operatorname{im} C$, so $z \oplus a$ lies outside the reference support and the amplitude ratio is zero. If the test succeeds, $z \oplus a = C(u \oplus d)$, and Eq. (58) gives the ratio as a product of factor ratios. Sampled configurations have nonzero amplitude, so the denominators are nonzero. Binary matrix-vector products require $O(nk)$ operations, while latent sampling and factor-ratio evaluation require $O(k)$ operations. Including the output bitstrings, both reference sampling and ratio evaluation require at most $O(nk + n)$ elementary operations after polynomial preprocessing, assuming bounded arithmetic cost for evaluating the factors and their ratios.

To prepare the reference state, we extend the columns of C to an invertible matrix $M \in \mathbb{F}_2^{n\times n}$. We prepare the first k qubits in the factor states

$$|\alpha_\ell(c)\rangle = f_\ell(0;c)\,|0\rangle + f_\ell(1;c)\,|1\rangle\,, \tag{C2}$$

and initialize the remaining $n-k$ visible qubits in $|0\rangle$. A reversible linear circuit implementing M maps each basis state $|u, 0^{n-k}\rangle$ to $|Cu\rangle$ and therefore maps the product of the factor states and zero states to $|\psi_c\rangle$. Standard binary elimination implements this map using $O(n^2)$ CNOT gates with unrestricted connectivity and no additional ancillary qubits [43]. We then apply $H^{\otimes n}$ to prepare $|\phi(c)\rangle$. Efficient quantum deployment also requires efficient preparation of the single-qubit factor states.

Appendix D: Sixteen-qubit conditional switching construction

1. Cycle basis and reference state

We specialize the binary-linear encoder construction to the cycle space of the graph in Section V A. We use column vectors throughout, with all matrix operations over $\mathbb{F}_2$.

We order the sixteen edges lexicographically as

$$(01, 02, 03, 04, 05, 06, 12, 13, 14, 15, 16, 23, 24, 25, 26, 34). \tag{D1}$$

The six root edges $01, \ldots, 06$ form a spanning tree. The remaining edges form the chord set $\mathcal{C} = \{12, 13, 14, 15, 16, 23, 24, 25, 26, 34\}$. Each chord and its two root edges define a fundamental triangle.

We represent these ten triangles by a matrix $B \in \mathbb{F}_2^{10\times 16}$. Each row contains ones on its triangle's three edges and zeros elsewhere. The chord columns form the identity matrix, so B has rank ten. The connected graph has cycle rank $16-7+1 = 10$, and these rows therefore form a basis of its cycle space. The encoding matrix in Section V is $C = B^{\mathsf{T}}$.

The six independent vertex switches generate the group G of 64 switching masks. Each switch selects either zero or two edges of every triangle, so $Bg = 0$ for every $g \in G$. Since $\ker B$ has dimension $16 - 10 = 6$, it also contains 64 masks. Thus, $G = \ker B$.

For a vector of chord bits $r \in \mathbb{F}_2^{10}$, the edge configuration is $B^{\mathsf{T}}r$. We define the reference state as

$$|\psi_\beta\rangle = \sum_{r\in\mathbb{F}_2^{10}} a_\beta(r)\,|B^{\mathsf{T}}r\rangle\,, \tag{D2}$$

$$a_\beta(r) = c_\beta^{10-|r|}(-is_\beta)^{|r|}, \tag{D3}$$

where $|r|$ is the Hamming weight of r,

$$c_\beta = \cos(t_\beta/2), \qquad s_\beta = \sin(t_\beta/2), \qquad t_\beta = \frac{\pi}{2}\tanh(1.5\beta). \tag{D4}$$

The normalized rotation factors and injectivity of B^T ensure normalization, as established in Appendix C. Applying the Hadamard layer gives the encoded input $|\phi_\beta\rangle = H^{\otimes 16}|\psi_\beta\rangle$.

2. Classical sampling and amplitude ratios

We sample the reference distribution by drawing the ten chord bits independently, with probability s_β^2 of obtaining one, and returning $B^\mathsf{T}r$. The chord entries of this configuration equal r because the chord columns of B form the identity matrix. They therefore also allow us to recover the latent bits from any reference sample.

For a bit-flip mask m, we collect its ten chord entries into u and test whether $B^\mathsf{T}u = m$. If the test fails, m lies outside the cycle space, and flipping a supported reference configuration by m gives zero amplitude. If the test succeeds, the flip maps $B^\mathsf{T}r$ to $B^\mathsf{T}(r \oplus u)$.

For $\beta \neq 0$, both c_β and s_β are nonzero, so every cycle-space configuration has nonzero reference amplitude. The ratio for a mask within the cycle space is

$$\frac{a_\beta(r \oplus u)}{a_\beta(r)} = \left[-i\tan\left(\frac{t_\beta}{2}\right)\right]^{|r\oplus u|-|r|}. \tag{D5}$$

Masks outside the cycle space give zero ratio. At $\beta = 0$, the reference state is $|0\rangle^{\otimes 16}$, so the ratio equals one for $m = 0$ and zero otherwise. We evaluate ratios only at sampled configurations with nonzero reference amplitude. This procedure provides classical sampling and amplitude-ratio access without enumerating the full sixteen-qubit distribution.

3. Switching symmetry

For every switching mask $g \in G$, the relation $Bg = 0$ gives

$$g^\mathsf{T}B^\mathsf{T}r = (Bg)^\mathsf{T}r = 0. \tag{D6}$$

Every reference configuration therefore has even parity on the edges selected by g. Consequently,

$$Z_g|\psi_\beta\rangle = |\psi_\beta\rangle, \qquad X_g|\phi_\beta\rangle = |\phi_\beta\rangle. \tag{D7}$$

The IQP unitary commutes with X_g, so applying it preserves this eigenvalue. The output amplitudes at x and $x \oplus g$ are therefore equal, giving

$$q_{\boldsymbol{\theta}}(x \mid \beta) = q_{\boldsymbol{\theta}}(x \oplus g \mid \beta), \qquad g \in G. \tag{D8}$$

This equality holds for every condition and IQP parameter vector. The encoder thus enforces the six independent vertex-switching symmetries, independently of training accuracy.